Hyperspectral Pansharpening With Adaptive Feature Modulation-Based Detail Injection Network

Yunsong Li, *Member, IEEE*, Yuxuan Zheng, *Member, IEEE*, Jiaojiao Li, *Member, IEEE*, Rui Song, *Member, IEEE*, and Jocelyn Chanussot, *Fellow, IEEE*

***Abstract*— Recently, deep learning-based methodologies have attained unprecedented performance in hyperspectral (HS) pansharpening, which aims to improve the spatial quality of HS images (HSIs) by making use of details extracted from the high-resolution panchromatic (HR-PAN) image. However, it remains challenging to incorporate the details into the pansharpened image effectively, while alleviating the spectral distortion simultaneously. To tackle this problem, in this article, we propose an adaptive feature modulation-based detail injection network (AFM-DIN) for HS pansharpening, which mainly consists of four phases: high-frequency details generation of the HR-PAN image, multiscale feature extraction of the upsampled HSI, AFM-based detail injection, and reconstruction of the HR-HSI. First, a novel octave convolution unit is employed to decompose the HR-PAN image into high and low frequencies, and then merge the high-frequency features together to generate the comprehensive PAN-details. Second, the spatial and spectral separable 3-D convolution units with multiple kernel sizes are designed to extract multiscale features of the upsampled HSI in a computationally efficient manner. Subsequently, by taking the critical PAN-details as prior, the proposed AFM module is able to not only incorporate the detail information effectively, but also adjust the injected details adaptively to ensure the spectral fidelity. Finally, the anticipated HR-HSI is obtained through adding the upsampled HSI to the predicted HSI-details reconstructed from informative modulated features. Extensive comparison experiments with several state-of-the-arts conducted on simulated and real HS data sets demonstrate that our proposed AFM-DIN can achieve superior pansharpening accuracy in both spatial and spectral aspects.**

***Index Terms*— Adaptive feature modulation, detail injection, hyperspectral (HS) pansharpening, octave convolution, spatial and spectral separable 3-D convolution.**

Manuscript received 23 November 2021; revised 15 February 2022, 15 May 2022, and 10 August 2022; accepted 11 September 2022. Date of publication 15 September 2022; date of current version 29 September 2022. This work was supported in part by the National Key Research and Development Program of China under Grant 2018AAA0102702; in part by the National Nature Science Foundation of China under Grant 61901343, Grant 62101414, Grant 61801359, Grant 61701360, and Grant 61671383; in part by the Science and Technology on Space Intelligent Control Laboratory under Grant ZDSYS-2019-03; in part by the China Postdoctoral Science Special Foundation under Grant 2018T111019; in part by the Open Research Fund of CAS Key Laboratory of Spectral Imaging Technology under Grant LSIT201924W; in part by the Fundamental Research Funds for the Central Universities under Grant JBF220101; in part by the Innovation Fund of Xidian University under Grant 5001-20109215456; in part by the Government-Business-University-Research Cooperation Foundation between Wuhu and Xidian under Grant XWYCXY-012021002-HT; in part by the 111 Project under Grant B08038; and in part by the China Scholarship Council under Grant 202006960027. *(Corresponding authors: Yuxuan Zheng; Jiaojiao Li.)*

Yunsong Li, Yuxuan Zheng, and Rui Song are with the State Key Laboratory of Integrated Services Networks, School of Telecommunications Engineering, Xidian University, Xi'an 710071, China (e-mail: ysli@mail.xidian.edu.cn; yxzheng24@163.com; rsong@xidian.edu.cn).

Jiaojiao Li is with the State Key Laboratory of Integrated Services Networks, School of Telecommunications Engineering, Xidian University, Xi'an 710071, China, and also with the CAS Key Laboratory of Spectral Imaging Technology, Xi'an 710119, China (e-mail: jjli@xidian.edu.cn).

Jocelyn Chanussot is with LJK, Grenoble INP, Inria, CNRS, Université Grenoble Alpes, 38000 Grenoble, France (e-mail: jocelyn.chanussot@grenoble-inp.fr).

Digital Object Identifier 10.1109/TGRS.2022.3206880

I. Introduction

HYPERSPECTRAL (HS) imaging technology that is capable of acquiring images across a broad range of the spectrum has been increasingly employed in remote sensing applications. The captured HS image (HSI) with rich spectral details has shown its significance in various fields including, among others, unmixing, land-cover classification, target detection, and anomaly detection [1], [2], [3], [4], [5], [6]. However, due to the design trade-offs between spatial and spectral resolutions, the HS cameras commonly provide high spectral resolution images with the sacrifice of spatial resolution [7]. For example, the Hyperion sensor is able to collect HS data with a 10 nm spectral resolution in the 400–2500 nm wavelength range, but with a relative low spatial resolution of 30 m [8]. In contrast, the single-band panchromatic (PAN) image is usually characterized by a high spatial resolution but a low spectral resolution. Considering that the HSI with both high spectral and spatial resolutions is desirable for many application domains [9], an effective solution for enhancing the spatial resolution of HS data is to fuse a low spatial resolution (LR) HSI (LR-HSI) with a high spatial resolution (HR) PAN (HR-PAN) image, which is well known as HS pansharpening [10].

To produce the expected HR-HSI, various pansharpening approaches have been proposed over the last decades, which can be generally divided into classical and emerging methods [11]. Among classical methods, component substitution (CS) and multiresolution analysis (MRA) are two widely used representatives. The CS approaches are based on replacing the spatial information of an HSI with an histogram-matched version of the HR-PAN image in a transformed domain, which includes intensity-hue-saturation (IHS) [12], Gram–Schmidt adaptive (GSA) [13], and principal component analysis (PCA) [14]. The category of MRA approaches relies on injecting spatial details extracted from the HR-PAN image through a multiscale decomposition into the HSI, which involves such methods as high-pass filtering (HPF) [15], smoothing filter-based intensity modulation (SFIM) [16], generalized Laplacian pyramid with modulation transfer function

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:38:43 UTC from IEEE Xplore. Restrictions apply.

matched filter (MTF-GLP) [17], and MTF-GLP with high-pass modulation (MTF-GLP-HPM, abbreviated as MGH in the following) [18]. In general, it remains a major challenge for the existing CS and MRA methods to improve the spatial quality of HSIs while retaining as much spectral information as possible [19]. On the one hand, CS methods are able to obtain fusion results with high spatial quality, but may lead to serious spectral distortions. On the other hand, the fused images derived by MRA techniques can better preserve the spectral properties, but may present less sharpness and details than the results obtained by CS approaches. As an emerging branch of remote sensing image fusion technology in recent years, deep learning (DL) methodology, especially the convolutional neural network (CNN), has been applied to the pansharpening task and exhibited unprecedented performance [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36]. This success can be attributed to the strong capability of CNN in extracting hierarchical features and learning nonlinear mappings. However, there are still certain limitations in the existing CNN-based pansharpening methods. First, a number of previous approaches merely employ some off-the-shelf image super-resolution (SR) network architectures for the pansharpening task, without considering the domain-specific knowledge of pansharpening, which results in the lack of interpretability for pansharpening [37]. Second, the extraction of PAN-details depends heavily on the complex design of filtering schemes, which inevitably leads to limited exploration of spatial information and complicated implementation of pansharpening. In addition, the current CNN-based methods can hardly achieve a satisfactory balance between spectral and spatial quality of pansharpened images, since the details complementary to HSIs are not adequately learned and not properly injected.

To overcome the aforementioned limitations, we propose an adaptive feature modulation-based detail injection network (AFM-DIN), for HS pansharpening. To the best of our knowledge, this study is the first effort that implements adaptive incorporation of detail information through feature modulation (FM) within the detail injection framework for tackling the challenging HS pansharpening problem. Experiments on both simulated and real HS data sets substantiate the superior performance of our AFM-DIN over several state-of-the-art pansharpening methods. The main contributions of this article are as follows.

1) Following the general idea of detail injection framework for HS pansharpening, a new AFM-DIN method is presented, which has a clear interpretation in the pansharpening context. Taking the critical PAN-details as prior, the elaborately designed AFM unit with the gating mechanism enables not only to incorporate the detail information effectively, but also to adjust the injected details adaptively to alleviate the spectral distortion.
2) A novel octave convolution (OctConv) unit is introduced for high-frequency details generation of the PAN image. We employ three OctConv layers to first decompose the PAN features into high and low frequencies and then combine all the high-frequency features together to obtain the comprehensive detail information, which is generated directly in a high-frequency feature representation and is proven to be an effective prior for boosting the pansharpening accuracy.
3) To extract multiscale spatial-spectral features of the HSI efficiently, we creatively design the spatial and spectral separable 3-D convolution (Sep 3DConv) units with different kernel sizes, which can capture diverse HS features while decreasing the number of parameters simultaneously.

The remainder of this article is structured as follows. Section II reviews the detail injection framework and CNN-based pansharpening methods. Section III introduces the proposed AFM-DIN method. Extensive experiments and comprehensive analysis are presented in Section IV. Section V gives the conclusions of this article.

II. Related Works

A. Detail Injection Framework for HS Pansharpening

Let $\mathbf{P} \in \mathbb{R}^{M\times N}$ denotes the available HR-PAN image with $M \times N$ pixels. Let $\mathbf{H} \in \mathbb{R}^{m\times n\times B}$ be the observed LR-HSI with $m \times n$ pixels and B bands, $\widetilde{\mathbf{H}} \in \mathbb{R}^{M\times N\times B}$ be the upsampled version of HSI (UP-HSI), and $\widehat{\mathbf{H}} \in \mathbb{R}^{M\times N\times B}$ be the pansharpened HR-HSI ($m \ll M$, $n \ll N$), respectively.

Generally, the classical CS and MRA pansharpening methods both include two consecutive phases: 1) detail extraction and 2) detail injection into the upsampled HSI. These processes can be formulated in a detail injection framework as follows:

$$\widehat{\mathbf{H}}_b = \widetilde{\mathbf{H}}_b + \mathbf{G}_b \otimes \mathbf{D}, \quad b = 1, 2, \ldots, B \tag{1}$$

where b indicates the bth spectral band, and the symbol $\otimes$ denotes element-wise multiplication of the injection gains matrix $\mathbf{G}_b$ and the detail image $\mathbf{D}$.

In CS methods, the detail image $\mathbf{D}$ is usually given by

$$\mathbf{D} = \mathbf{P} - \mathbf{I}_{\mathrm{L}} \tag{2}$$

where $\mathbf{I}_{\mathrm{L}} \in \mathbb{R}^{M\times N}$ is the intensity component obtained by a linear combination of multiple bands of $\widetilde{\mathbf{H}}$

$$\mathbf{I}_{\mathrm{L}} = \sum_{b=1}^{B} \omega_b \widetilde{\mathbf{H}}_b \tag{3}$$

in which ω_b represents the bth weight. Thus, algorithms in the CS family mainly differ in their ways to get the weights ω_b and the injection coefficients $\mathbf{G}_b$. While in MRA methods, $\mathbf{D}$ is generally defined as

$$\mathbf{D} = \mathbf{P} - \mathbf{P}_{\mathrm{L}} \tag{4}$$

where $\mathbf{P}_{\mathrm{L}}$ denotes a low-pass version of the HR-PAN image $\mathbf{P}$, which is often derived through an iterative decomposition of $\mathbf{P}$. Accordingly, methods belonging to the MRA category can differ from each other in the way of obtaining the image $\mathbf{P}_{\mathrm{L}}$ and calculating the injection gains $\mathbf{G}_b$.

Although CS and MRA approaches have been widely investigated and achieved great developments in recent decades, the existing pansharpening techniques belonging to these two

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:38:43 UTC from IEEE Xplore. Restrictions apply.

classes can easily suffer from limitations such as poor preservation of spectral and spatial information. This is mainly due to the utilization of linear models in traditional CS and MRA frameworks, which may result in insufficient exploitation and injection of details [23], [25].

B. CNN-Based Pansharpening Methods

In the last few years, inspired by the impressive results obtained by DL technology in many applicative fields especially in single image SR [38], [39], [40], various CNNs with the high nonlinearity have been employed to solve the problem of the linear pansharpening models discussed above. Following the super-resolution convolutional neural network (SRCNN) [38] structure, Masi *et al.* [20] proposed a CNN-based pansharpening method, where the interpolated multispectral image (MSI) and HR-PAN image were stacked together as the input, and a three-layer CNN was adopted to learn the mapping between the input and the target HR-MSI. Aiming at preserving spatial and spectral information simultaneously, a pansharpening network called PanNet was presented in [21]. By training network parameters in the high-pass domain rather than the image domain, the PanNet architecture demonstrated a better ability of generalization to new satellites. Compared with PanNet, the deep multiscale detail network (DMDNet) [22] with grouped multiscale dilated convolutions can obtain larger receptive fields. Afterward, some detail injection framework-based CNN pansharpening methods have been developed [23], [24], [25]. For example, He *et al.* [23] designed two detail injection-based CNNs, i.e., DiCNN1 and DiCNN2, for learning the MSI details in an end-to-end manner. Similarly, a detail-based deep Laplacian pansharpening (DDLPS) [24] method was proposed for the fusion of HS and PAN images, where the Laplacian pyramid super-resolution network (LapSRN) [40] was used to perform upsampling of the LR-HSI. Motivated by the fact that a strong spectral-prediction capability is desired for CNN-based HS pansharpening methods, He *et al.* [26] developed two specific models, namely hyperspectral pansharpening via spectrally predictive CNN (HyperPNN)1 and HyperPNN2, to strengthen the spectral prediction ability of CNNs. To tackle the challenges of reconstructing HR-HSIs beyond the training scale and guaranteeing spectral fidelity at any spatial resolutions, a CNN-based method for arbitrary resolution HS pansharpening was presented in [27], which mainly comprises a multilevel enhancement subnetwork and a rescaling subnetwork. More recently, attention mechanisms have been introduced to DL-based pansharpening models for improving the representational power of CNNs and further boosting the accuracy of fusion [28], [29], [30], [31], [32]. Among those, Zheng *et al.* [29] proposed a dual-attention residual network (DARN) to simultaneously learn more informative channel-wise and spatial-domain features for HS pansharpening. To alleviate the problems of existing pansharpening methods such as spectral distortion and detail loss, Liu *et al.* [31] proposed a hybrid attention mechanism-based residual neural network named HARNN for the pansharpening task, which was able to select more valuable spectral and spatial features for improving the fusion efficiency. Guan and Lam [32] developed a multistage dual-attention guided fusion network (MDA-Net) for HS pansharpening, where a dual-attention guided fusion block (DAFB) with spectral and spatial attention mechanisms is utilized to fuse the features extracted from three streams in multiple stages. Apart from the aforementioned CNN-based methods which were all trained in a supervised fashion, a few attempts that using the unsupervised models have recently been made for pansharpening. An unsupervised pansharpening method based on the self-attention mechanism was proposed in [33], where the detail extraction and injection functions were predicted according to the attention representations. Luo *et al.* [34] designed an unsupervised CNN-based pansharpening model with an iterative architecture to jointly extract and fuse the features derived from the input. In [35], an unsupervised framework with registration learning for pan-sharpening was proposed, where the inherent misalignment between input images was considered. Nie *et al.* [36] proposed an unsupervised HS pansharpening method by ratio estimation (RE) and residual attention network (RE-RANet), where the spatial and spectral information are derived from the original PAN and HSIs rather than the reduced-resolution images. Despite recent advances obtained by CNN-based approaches in pansharpening, it is still challenging to incorporate the spatial detail information extracted from the PAN image into the HS bands effectively, and to alleviate the spectral distortion of the pansharpened image simultaneously due to the limited capacity of detail extraction and injection.

C. Octave Convolution

To reduce the spatial redundancy and improve the network efficiency, Chen *et al.* [41] proposed the octave convolution, which factorizes the vanilla convolutional features into high- and low-frequency components. Specifically, let $X \in \mathbb{R}^{h\times w\times c}$ denotes the input feature tensor, which consists of c channels with spatial size of $h \times w$. With a ratio $\alpha \in [0, 1]$, X can be factorized into $X = \{X_{\mathrm{H}}, X_{\mathrm{L}}\}$ along the channel dimension, where the high-frequency feature maps $X_{\mathrm{H}} \in \mathbb{R}^{h\times w\times(1-\alpha)c}$ capture fine details and the low-frequency ones $X_{\mathrm{L}} \in \mathbb{R}^{(h/2)\times(w/2)\times\alpha c}$ change slower in the spatial dimensions. Then, the octave convolution can run directly on $\{X_{\mathrm{H}}, X_{\mathrm{L}}\}$ to not only process the low and high frequencies in their corresponding frequency tensor, but also enable efficient inter-frequency communication. Let $Y = \{Y_{\mathrm{H}}, Y_{\mathrm{L}}\}$ be the output feature tensor. The high-frequency feature maps Y_{H} and the low-frequency feature maps Y_{L} can be, respectively, expressed as

$$\begin{aligned} Y_{\mathrm{H}} &= f(X_{\mathrm{H}}; W_{\mathrm{H}\to\mathrm{H}}) + \mathrm{Upsample}(f(X_{\mathrm{L}}; W_{\mathrm{L}\to\mathrm{H}})) \\ Y_{\mathrm{L}} &= f(X_{\mathrm{L}}; W_{\mathrm{L}\to\mathrm{L}}) + f(\mathrm{AvgPool}(X_{\mathrm{H}}); W_{\mathrm{H}\to\mathrm{L}}) \end{aligned} \tag{5}$$

where $f(X; W)$ represents a convolution with parameters W, $\mathrm{Upsample}(\cdot)$ denotes an upsampling operation by nearest interpolation, and $\mathrm{AvgPool}(\cdot)$ is an average pooling operation. $W_{\mathrm{H}\to\mathrm{H}}$ and $W_{\mathrm{L}\to\mathrm{L}}$ denote intrafrequency update, while $W_{\mathrm{L}\to\mathrm{H}}$ and $W_{\mathrm{H}\to\mathrm{L}}$ denote interfrequency communication.

The detailed design of the octave convolution is shown in Fig. 1. The octave convolution operator consists of four paths

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:38:43 UTC from IEEE Xplore. Restrictions apply.

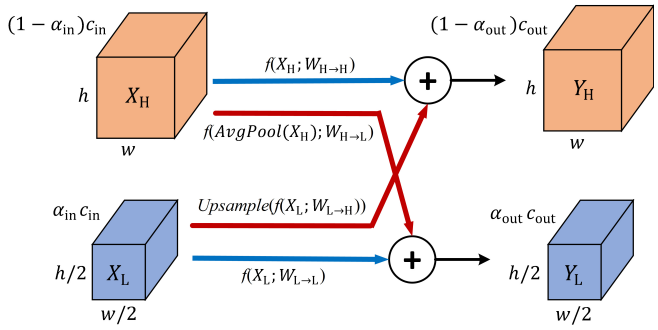


Fig. 1. Detailed design of the octave convolution.

corresponding to the four terms in (5). Two blue arrows correspond to information updating, and two red arrows accomplish information communication between the two frequencies. As illustrated in Fig. 1, the output high-frequency feature map Y_H is obtained by adding the convolution of the input high-frequency part X_H and the upsampled low-frequency part. The output low-frequency feature map Y_L is generated through adding the convolved low-frequency part X_L and the pooled original high-frequency part.

III. Proposed Method

In this section, we illustrate the AFM-DIN method in detail. First, we present an overview of our AFM-DIN. Then, we describe the four key parts of AFM-DIN concretely, including high-frequency details generation of the HR-PAN image via OctConv unit, multiscale feature extraction of the UP-HSI through Sep 3DConv unit, adaptive FM-based detail injection, and reconstruction of the HR-HSI.

A. Overview of Network Architecture

The overall architecture of our proposed AFM-DIN method is illustrated in Fig. 2. As shown in Fig. 2, the AFM-DIN approach is mainly composed of four phases: high-frequency details generation, multiscale feature extraction, adaptive FM, and high-resolution image reconstruction.

In the first phase, the HR-PAN image $\mathbf{P}$ is fed into the OctConv unit to generate the high-frequency PAN-details F_{PD}

$$F_{PD} = f_{OC}(\mathbf{P}) \tag{6}$$

where f_{OC} denotes the OctConv unit mainly comprising three OctConv layers, which will be detailed in Section III-B. F_{PD} represents the feature maps of PAN-details, which are forwarded to the corresponding FM blocks for adaptive incorporation of detail information. To extract multiscale spatial-spectral features of the UP-HSI $\widetilde{\mathbf{H}}$ in a computationally efficient manner, we creatively design the Sep 3DConv units with different kernel sizes in the second phase. Thus, the HS features F_{HS} can be obtained by

$$F_{HS} = f_{1\times1}\left(\left[f_{SC}^{k_1}\left(\widetilde{\mathbf{H}}\right), f_{SC}^{k_2}\left(\widetilde{\mathbf{H}}\right), f_{SC}^{k_3}\left(\widetilde{\mathbf{H}}\right)\right]\right) \tag{7}$$

where $f_{SC}^{k_1}(\cdot)$, $f_{SC}^{k_2}(\cdot)$, and $f_{SC}^{k_3}(\cdot)$ represent the three Sep 3DConv units with kernel sizes of k_1, k_2, and k_3, respectively, $[\cdot]$ is the concatenation operation, and $f_{1\times1}(\cdot)$ denotes the function of the 1×1 convolution (Conv) layer. Subsequently, the generated F_{HS} is sent to the AFM module for FM conditioned on F_{PD}, where several FM blocks equipped with AFM units are cascaded. Furthermore, short and long skip connections are adopted in each FM block and across all the K FM blocks, respectively, which not only contribute to enhancing the information flow but also help to ease the training burden. Consequently, the FM process in the third phase can be expressed as

$$F_M = F_{HS} + f_{FM}^{K}\left(f_{FM}^{K-1}\left(\cdots f_{FM}^{1}(F_{HS})\cdots\right)\right) \tag{8}$$

where $f_{FM}^{K}(\cdot)$ represents the function for the Kth FM block which integrates two AFM units into the residual block to perform AFM on F_{HS}, and F_M denotes the modulated features. After obtaining the high informative features F_M, we exploit a detail reconstruction (DR) block for high-resolution image reconstruction in the last phase, which can be formulated as

$$\widehat{\mathbf{H}} = \widetilde{\mathbf{H}} + \widehat{\mathbf{H}}_D = \widetilde{\mathbf{H}} + f_{DR}(F_M) \tag{9}$$

where $f_{DR}(\cdot)$ denotes the function of the DR block, and $\widehat{\mathbf{H}}_D = f_{DR}(F_M)$ is the estimated details of the HSI (HSI-details). Finally, $\widehat{\mathbf{H}}_D$ is added to the UP-HSI $\widetilde{\mathbf{H}}$ to produce the expected pansharpened image $\widehat{\mathbf{H}}$.

As suggested in previous studies [29], [42], [43], we adopt L_1 loss function to minimize the difference between the pansharpened image $\widehat{\mathbf{H}}$ and the ground-truth (GT) HR-HSI $\mathbf{H}_G$, which is defined as

$$L_1(\Theta) = \frac{1}{N_p}\sum_{i=1}^{N_p}\|\widehat{\mathbf{H}}^i - \mathbf{H}_G^i\|_1 \tag{10}$$

where N_p is the total number of training samples, i denotes the index of samples, and Θ represents the parameters of our AFM-DIN.

B. High-Frequency Details Generation of the HR-PAN Image via OctConv Unit

HS pansharpening refers to the process of improving the spatial quality of HSIs by utilizing details extracted from the HR-PAN image [44]. Therefore, the high-frequency details of the PAN image play a vital role in pansharpening. However, in most existing pansharpening methods, the extraction of PAN-details still relies heavily on the complicated design of spatial filters, which often fails to fully explore the spatial detail information of the PAN image [45]. Considering this limitation, we introduce a novel OctConv unit to capture more comprehensive information of the high-frequency PAN-details efficiently, which enables extracting details directly from the high-frequency feature maps, thus circumventing the problem of complicated design of filtering schemes and limited exploitation of detail information.

The octave convolution proposed in [41] assumes that the output features of a Conv layer can be decomposed into high- and low-frequency groups, which, respectively, represent fine details and global structures. Thus, the two groups could be processed at their corresponding frequency with effective information update and exchange. Based on this, we develop a novel OctConv unit which employs three octave convolution layers to first factorize the PAN feature maps into high and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:38:43 UTC from IEEE Xplore. Restrictions apply.

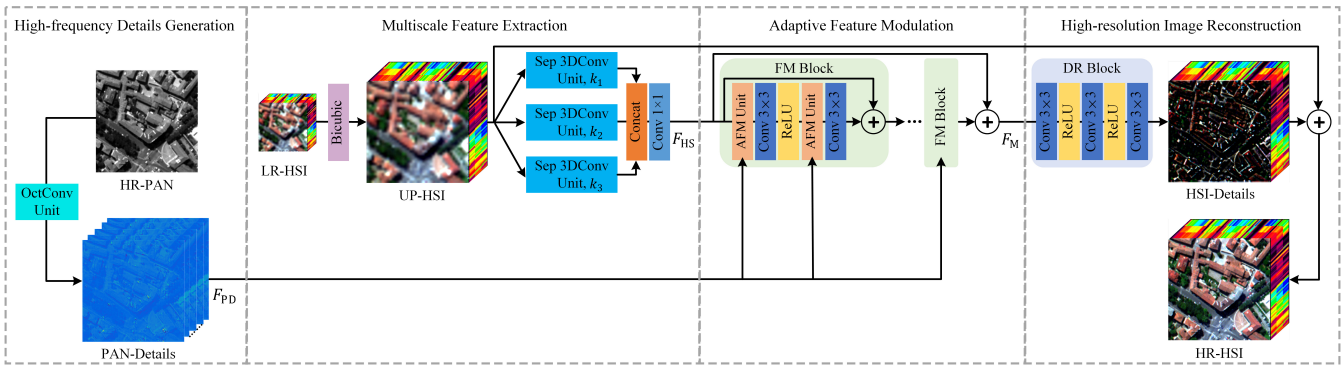


Fig. 2. Overall architecture of the proposed AFM-DIN method for HS pansharpening.

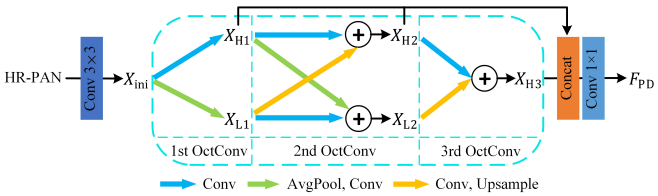


Fig. 3. Detailed structure of the OctConv unit.

low frequencies, and then combine the high-frequency features together to generate the elaborate feature representation of PAN-details.

The structure of our OctConv unit is detailed in Fig. 3, which mainly consists of three parts: initial feature extraction, high-frequency feature generation, and hierarchical feature fusion. As depicted in Fig. 3, given the input HR-PAN image $\mathbf{P}$, the initial feature representation $X_{\text{ini}} \in \mathbb{R}^{h\times w\times c}$ can be computed by

$$X_{\text{ini}} = f_{3\times 3}(\mathbf{P}) \tag{11}$$

where $f_{3\times 3}(\cdot)$ denotes the 3×3 Conv layer.

Then, the first OctConv layer is employed to decompose the initial feature maps X_{ini} into $\{X_{\text{H1}}, X_{\text{L1}}\}$ along the channel dimension, where $X_{\text{H1}} \in \mathbb{R}^{h\times w\times(1-\alpha)c}$ and $X_{\text{L1}} \in \mathbb{R}^{(h/2)\times(w/2)\times\alpha c}$ represent, respectively, the high-frequency group and the low-frequency group, and $\alpha \in [0, 1]$ is the ratio of channels assigned to the low-frequency part X_{L1}. This decomposing process can be formulated as

$$\begin{aligned} X_{\text{H1}} &= f_{3\times 3}(X_{\text{ini}}) \\ X_{\text{L1}} &= f_{3\times 3}(\text{AvgPool}(X_{\text{ini}})) \end{aligned} \tag{12}$$

where AvgPool$(\cdot)$ denotes average pooling operation. Subsequently, intrafrequency update and interfrequency communication are performed simultaneously on X_{H1} and X_{L1} in the middle layer. The output of the second OctConv layer, denoted as X_{H2} and X_{L2}, can be obtained by

$$\begin{aligned} X_{\text{H2}} &= f_{3\times 3}(X_{\text{H1}}) + \text{Upsample}(f_{3\times 3}(X_{\text{L1}})) \\ X_{\text{L2}} &= f_{3\times 3}(X_{\text{L1}}) + f_{3\times 3}(\text{AvgPool}(X_{\text{H1}})) \end{aligned} \tag{13}$$

where Upsample$(\cdot)$ indicates upsampling operation. In this way, our OctConv unit is able to not only handle the high- and low-frequency components in their corresponding frequency effectively, but also exchange interfrequency information efficiently. In the following third OctConv layer, the high-frequency group X_{H2} and the low-frequency group X_{L2} are processed to realize the same spatial resolution and then added up to generate a single high-frequency output X_{H3}

$$X_{\text{H3}} = f_{3\times 3}(X_{\text{H2}}) + \text{Upsample}(f_{3\times 3}(X_{\text{L2}})). \tag{14}$$

Finally, to obtain rich detail information for HS pansharpening, we adopt the hierarchical feature fusion to combine the high-frequency feature maps of each OctConv layer, which can be formulated as

$$F_{\text{PD}} = f_{1\times 1}([X_{\text{H1}}, X_{\text{H2}}, X_{\text{H3}}]) \tag{15}$$

where the meanings of $[\cdot]$ and $f_{1\times 1}(\cdot)$ are the same as those in (7). F_{PD} is the generated comprehensive feature representation of the high-frequency PAN-details.

C. Multiscale Feature Extraction of the UP-HSI Through Sep 3DConv Unit

Typically, an HSI is presented in a 3-D data cube which contains two spatial dimensions and one spectral dimension. Since 3DConv is able to explore both spatial context and spectral correlation at the same time, employing 3DConv for feature extraction of HS data has become a research focus [46], [47], [48]. However, compared with utilizing 2D-CNN alone, simply adopting 3DConv instead of 2DConv can lead to a significant increase in network parameters and computational complexity. This is due to the fact that 3DConv is implemented by convolving a 3-D kernel with the input. To extract spatial-spectral features of the HSI in different scales efficiently, we design the spatial and spectral Sep 3DConv units with multiple kernel sizes, which allows our network to capture multiscale HS feature representations with decreased complexity.

The detailed architecture of our Sep 3DConv unit is illustrated in Fig. 4, which is composed of 3DConv, rectified linear unit (ReLU), and reshape operations. Specifically, a 3DConv layer with the filter $k \times k \times 1$ is used to first extract the spatial features of the input HSI, and then a 3DConv with filter size of $1 \times 1 \times$ k is adopted to explore the features between spectra. That is to say, the original filter $k \times k \times k$ is separated into two forms: $k \times k \times 1$ and $1 \times 1 \times$ k. Obviously, by separating the filter utilized in standard 3DConv, the number of model parameters can be greatly reduced. This factorization is similar in spirit to the separable 3D-CNN proposed in [49], except that we apply the second Conv operation to the spectral dimension rather than the temporal

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:38:43 UTC from IEEE Xplore. Restrictions apply.



Fig. 4. Detailed architecture of the Sep 3DConv unit.

dimension. In our Sep 3DConv unit, the two Conv layers are both activated by the ReLU functions. After the above operations, the obtained spatial-spectral features are reshaped from the size of $H \times W \times D \times C$ to $H \times W \times (D \times C)$ for further processing, where H, W and D are the height, width, and depth of the feature maps, and C denotes the number of feature maps. Thus, the output of one Sep 3DConv unit with kernel size of k, denoted as F_{SC}^{k}, can be formulated as

$$F_{\mathrm{SC}}^{k} = \mathrm{Reshape}\Big(\delta\Big(f_{1\times1\times k}\Big(\delta\Big(f_{k\times k\times1}\Big(\widetilde{\mathbf{H}}\Big)\Big)\Big)\Big)\Big) \tag{16}$$

where $f_{k\times k\times1}(\cdot)$ and $f_{1\times1\times k}(\cdot)$ represent the spatial Conv with filter shape $k \times k \times 1$ and the spectral Conv with filter shape $1 \times 1 \times k$, respectively. $\delta(\cdot)$ is the ReLU function, and Reshape$(\cdot)$ denotes the reshape operation.

As displayed in Fig. 2, in the second phase, the LR-HSI $\mathbf{H}$ is first upscaled to the UP-HSI $\widetilde{\mathbf{H}}$ using bicubic interpolation, which is widely preferred in pansharpening methods [50]. Next, $\widetilde{\mathbf{H}}$ is inputted to three branches for multiscale feature extraction, each of which is composed of one Sep 3DConv unit. It is worth noting that the three Sep 3DConv units are with different kernel sizes of $k_1 = 3$, $k_2 = 5$, and $k_3 = 7$, respectively. Finally, the multiscale HS features F_{HS} can be derived from (7), where the concatenation and the convolution operations are employed successively.

D. Adaptive FM-Based Detail Injection

One of the central problems in HS pansharpening is the difficulty of obtaining a satisfactory balance between spectral and spatial quality due to insufficient learning and injection of detail information [23], [25], as discussed in Section I. To tackle this problem, we propose a new AFM module to achieve effective and adaptive incorporation of PAN details by cascading several FM blocks equipped with elaborately designed AFM units. Particularly, we introduce the gating mechanism into each AFM unit that not only adjusts the modulation parameters flexibly, but also captures more important detail information adaptively.

Now we take the first AFM unit as an example to illustrate its architecture, which is shown in Fig. 5. Given the PAN details' features $F_{\mathrm{PD}} \in \mathbb{R}^{h\times w\times c'}$, we first apply a mapping function $f_M(\cdot)$ which consists of two parallel three-layer 1×1 convolutions connected by two ReLU layers to generate the intermediate parameters α' and β'

$$(\alpha', \beta') = f_M(F_{\mathrm{PD}}). \tag{17}$$

Here, a neural network is adopted as the mapping function so as to be optimized end-to-end. Then, to achieve adaptive control of detail information, the gating mechanism with sigmoid functions is used to obtain the scaling tensor α and the shifting tensor β

$$\begin{aligned} \alpha &= \alpha' \otimes \sigma(\alpha') \\ \beta &= \beta' \otimes \sigma(\beta') \end{aligned} \tag{18}$$

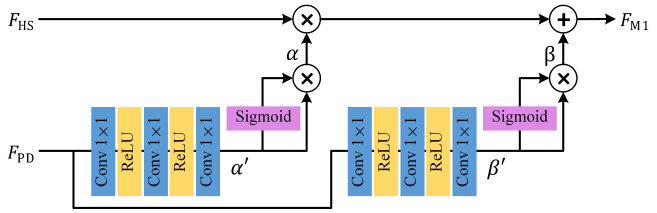


Fig. 5. Architecture of the proposed AFM unit.

where $\sigma(\cdot)$ refers to the sigmoid function which squashes values to the range between 0 and 1, and $\otimes$ denotes the element-wise multiplication operation. After obtaining the affine transformation parameters (α, β) from the PAN details prior, the AFM unit modulates the HS features $F_{\mathrm{HS}} \in \mathbb{R}^{h\times w\times c'}$ adaptively to incorporate the PAN detail information into the HS features effectively. This process can be expressed as

$$F_{\mathrm{M1}} = f_{\mathrm{AFM}}(F_{\mathrm{HS}}) = \boldsymbol{\alpha} \otimes F_{\mathrm{HS}} \oplus \boldsymbol{\beta} \tag{19}$$

where $\oplus$ denotes the element-wise addition. $f_{\mathrm{AFM}}(\cdot)$ indicates the function of our AFM unit, and F_{M1} is the modulated feature representation of the first AFM unit. With the above process, our AFM unit is able to maintain crucial PAN details adaptively, and modulate HS features for boosting pansharpening performance accordingly.

As shown in Fig. 2, we apply AFM to F_{HS} by stacking multiple FM blocks in the AFM-DIN. Thus, based on (8), the high informative modulated features F_{M} are acquired in the third phase, where the critical PAN detail information is injected adaptively and effectively.

E. Reconstruction of the HR-HSI

With respect to the phase of HR-HSI reconstruction, we first develop a DR block to estimate the missing HSI details from the modulated features F_{M}. The DR block is composed of three adjacent 3×3 Conv layers, among which two ReLU activation functions are inserted. Finally, following (9), the desired HR-HSI can be reconstructed by adding the estimated HSI details to the UP-HSI.

IV. Experiments and Analysis

In this section, we first introduce the experimental data sets, implementation details and evaluation metrics. Then, we analyze the effectiveness of our model through ablation study. Subsequently, we compare the proposed AFM-DIN with several state-of-the-art pansharpening methods in terms of quantitative and visual assessments. Finally, we make the analysis of model complexity for CNN-based methods.

A. Data Sets and Implementation Details

To evaluate the performance of the proposed AFM-DIN method for HS pansharpening, three simulated data sets, i.e., Botswana, Pavia Center, and Washington DC Mall data sets and one real data set named Los Angeles were utilized in our experiments.

1) *Botswana*: The first data set named Botswana was collected by the Hyperion sensor on the National Aeronautics and Space Administration (NASA) Earth Observing One (EO-1) satellite over the Okavango

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:38:43 UTC from IEEE Xplore. Restrictions apply.

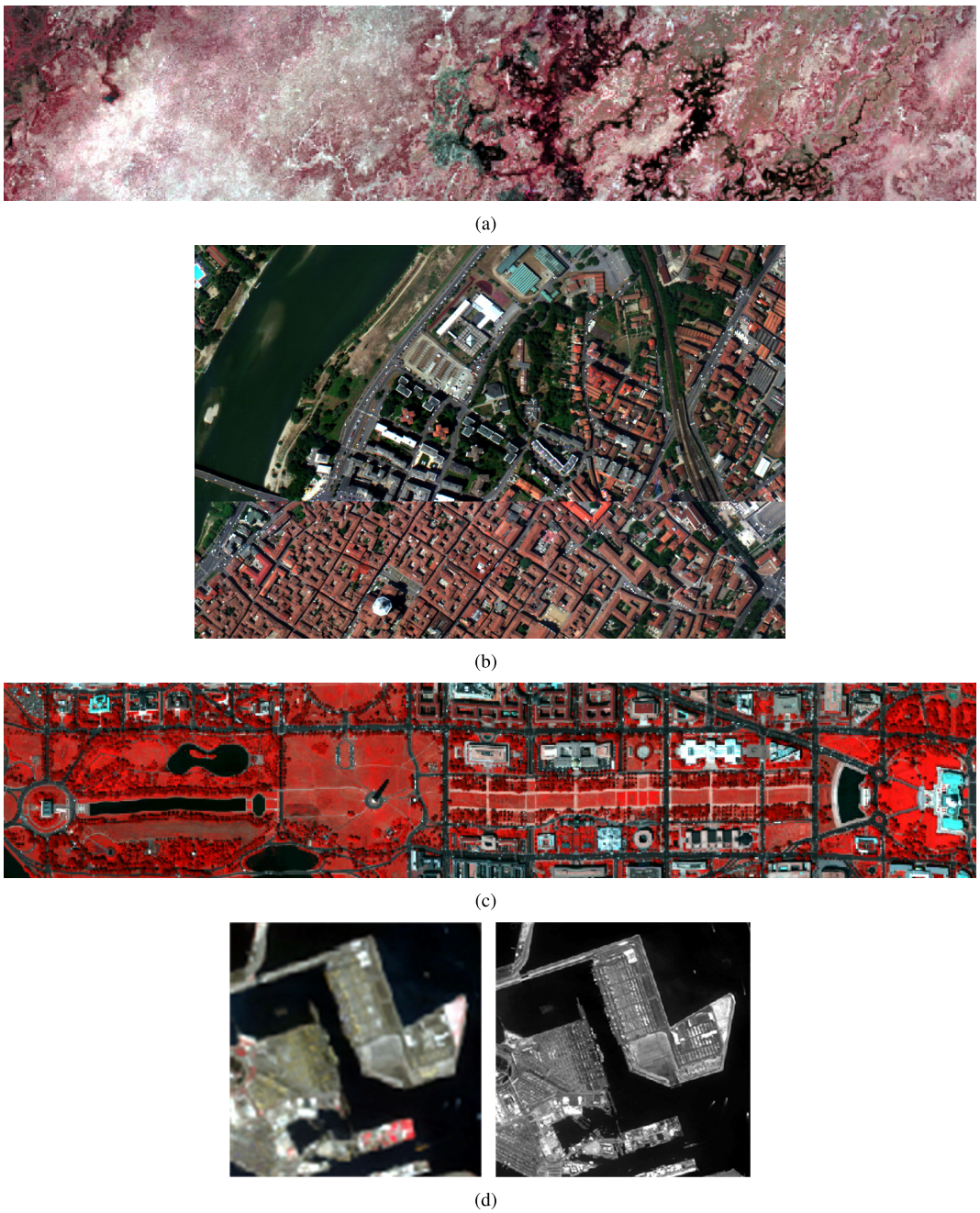


Fig. 6. Pseudocolor images of four experimental data sets. Note that the scenes were rotated 90° counterclockwise for display. (a) Botswana (Red: band 29, Green: band 20 and Blue: band 12). (b) Pavia Center (Red: band 60, Green: band 30 and Blue: band 10). (c) Washington DC Mall (Red: band 60, Green: band 27 and Blue: band 17). (d) Los Angeles (HSI (left): Red: band 29, Green: band 20 and Blue: band 12; PAN image (right).

Delta, Botswana. The image originally had 242 bands covering the spectral range of 400–2500 nm. After removing the corrupted bands, there remained 145 bands for study. In our experimental design, only the top left part of the image with a spatial size of 1200 × 240 was intercepted and used, as shown in Fig. 6(a). The selected part was then divided into 20 nonoverlapping cubic patches of size 120 × 120 × 145, which served as the GT images. The LR-HSIs with the size of 30 × 30 × 145 were simulated via applying a Gaussian blur and down-sampling in spatial dimensions on the reference HR-HSIs, i.e., the scaling factor was 4. The HR-PAN images with 120 × 120 pixels were generated by averaging the visible spectral range bands of GT images. In the training phase, 14 pairs (70%) of LR-HSIs and HR-PAN images from the Botswana data set were randomly selected. The remaining six patches were used in the testing stage.

2) *Pavia Center*: The second data set was acquired by the reflective optics system imaging spectrometer (ROSIS) sensor over the center of Pavia, Italy. The black strip samples and noisy bands of the original data were discarded, resulting in an image with 1096 × 715 pixels and 102 bands. Fig. 6(b) shows the pseudocolor image of the selected area for experimentation, which has 960 × 640 pixels. This area was partitioned into 24 image patches of size 160 × 160 × 102 without overlapping, which were then used to synthesize LR-HSIs and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:38:43 UTC from IEEE Xplore. Restrictions apply.

HR-PAN images. Accordingly, the size of LR-HSI was 40×40 with a spatial resolution of 5.2 m, and the HR-PAN image had 160×160 pixels with a spatial resolution of 1.3 m. In the Pavia Center dataset, 17 patches (about 70%) were chosen randomly as the training set, and the rest seven images were used for testing.

3) *Washington DC Mall*: The third data set was captured over the Washington DC Mall by the hyperspectral digital imagery collection experiment (HYDICE) sensor, containing 1280×307 pixels and 191 bands in the 400 to 2400 nm region of visible and infrared spectrum. The top right 1200×240-pixel-size image was utilized for the experimental design, as displayed in Fig. 6(c). We first separated it into 20 patches of size $120 \times 120 \times 191$ without intersection, and then simulated the LR-HSIs through Gaussian blurring and downsampling by a decimation factor of 4. The spatial sizes of obtained LR-HSIs and HR-PAN images were 30×30 and 120×120, respectively. The training set consisted of 14 patches (70%) from the Washington DC Mall data set, and the remainder comprised the testing set.
4) *Los Angeles*: The last data set was taken over a port in the Los Angeles city by Hyperion and advanced land imager (ALI) sensors mounted on the EO-1 satellite, which is a real HS data set. The HSI provided by Hyperion included 242 bands with a spatial resolution of 30 m, and the PAN image of the same scene provided by ALI had a spatial resolution of 10 m [29], [45], [51]. Uncalibrated and noisy spectral bands were discarded, and 145 bands were processed for the experimental design. The spatial size of the HSI and the PAN image used for experimentation were 120×120 and 360×360, as exhibited in Fig. 6(d). Note that the HSI is zoomed in three times here for better display. In the experiments, we segmented them into 16 nonoverlapping image pairs with the size 30×30 and 90×90 for the HS and PAN images, respectively. In the Los Angeles data set, 11 pairs (about 70%) were randomly selected for training, and the remaining five patches were utilized for testing. In the case of reduced-resolution evaluations, taking the original HSIs as the GT images, the available HSIs and PAN images were degraded using a Gaussian filter with a factor of 3. For the full-resolution assessments, pansharpening was performed at the original scale.

The experiments were conducted on a workstation with Ubuntu 16.04 operating system and Tensorflow library. Both training and testing of our AFM-DIN were carried out on a single NVIDIA GTX 1080Ti graphic processing unit (GPU). The spatial sizes of the training patches were 32×32 and 16×16 for the first three simulated data sets and for the last Los Angeles data set, respectively. We made use of the Adam optimizer with a learning rate of 10^{-4} to minimize the L_1 loss function. The batch size and the number of training epochs were empirically set to 16 and 2×10^3, respectively.

B. Evaluation Metrics

To quantitatively evaluate the quality of the pansharpened images for the simulated data sets, four popular full-reference metrics [10] are adopted in this article, including cross correlation (CC), spectral angle mapper (SAM), root mean squared error (RMSE) and erreur relative globale adimensionnelle de synthèse (ERGAS).

1) *CC*: The CC is a spatial index which characterizes the spatial distortion of the pansharpened image $\widehat{\mathbf{H}}$. For the bth band of $\widehat{\mathbf{H}}$ and the GT $\mathbf{H}_{\mathrm{G}}$, CC is defined as

$$\mathrm{CC} = \frac{\sum_{j=1}^{\mathrm{MN}} \left(X_j - \mu\right)\left(\widehat{X}_j - \hat{\mu}\right)}{\sqrt{\sum_{j=1}^{\mathrm{MN}} \left(X_j - \mu\right)^2 \sum_{j=1}^{\mathrm{MN}} \left(\widehat{X}_j - \hat{\mu}\right)^2}} \tag{20}$$

where X_j and $\widehat{X}_j$ denote the jth element of $\mathbf{H}_{\mathrm{G}}^b$ and $\widehat{\mathbf{H}}^b$, respectively. μ and $\hat{\mu}$ are the sample means of $\mathbf{H}_{\mathrm{G}}^b$ and $\widehat{\mathbf{H}}^b$. The CC has an optimal value of 1.

2) *SAM*: The SAM measures the spectral quality of the pansharpening result. For each pixel (p, q) of $\mathbf{H}_{\mathrm{G}}$ and $\widehat{\mathbf{H}}$, SAM is calculated as

$$\mathrm{SAM} = \arccos\left(\frac{\left\langle \mathbf{H}_{\mathrm{G}}^{(p,q)}, \widehat{\mathbf{H}}^{(p,q)} \right\rangle}{\left\| \mathbf{H}_{\mathrm{G}}^{(p,q)} \right\|_2 \left\| \widehat{\mathbf{H}}^{(p,q)} \right\|_2}\right) \tag{21}$$

where $\langle \cdot, \cdot \rangle$ denotes the dot product, and $\| \cdot \|_2$ is the L_2 norm. The overall SAM can be obtained by averaging the values computed for all the pixels. A smaller SAM value indicates less spectral distortion.

3) *RMSE*: The RMSE, which measures the L_2 error between $\widehat{\mathbf{H}}$ and $\mathbf{H}_{\mathrm{G}}$, is defined as

$$\mathrm{RMSE} = \frac{\left\| \widehat{\mathbf{H}} - \mathbf{H}_{\mathrm{G}} \right\|_F}{\sqrt{\mathrm{MNB}}} \tag{22}$$

where $\|\widehat{\mathbf{H}} - \mathbf{H}_{\mathrm{G}}\|_F = (\mathrm{trace}[(\widehat{\mathbf{H}} - \mathbf{H}_{\mathrm{G}})^T (\widehat{\mathbf{H}} - \mathbf{H}_{\mathrm{G}})])^{1/2}$ represents the Frobenius norm of $(\widehat{\mathbf{H}} - \mathbf{H}_{\mathrm{G}})$. The smaller the RMSE, the better the quality of $\widehat{\mathbf{H}}$.

4) *ERGAS*: The ERGAS with the ideal value of 0 provides a global evaluation of the pansharpened image, which is formulated as

$$\mathrm{ERGAS} = 100 \frac{M}{m} \sqrt{\frac{1}{B} \sum\nolimits_{b=1}^{B} \left[\frac{\mathrm{RMSE}_b}{\mu\left(\mathbf{H}_{\mathrm{G}}^b\right)}\right]^2} \tag{23}$$

where $\mathrm{RMSE}_b = (1/(\mathrm{MN})^{1/2}) \|\widehat{\mathbf{H}}^b - \mathbf{H}_{\mathrm{G}}^b\|_F$, and $\mu(\mathbf{H}_{\mathrm{G}}^b)$ is the mean of the bth band image of $\mathbf{H}_{\mathrm{G}}$.

Besides the abovementioned full-reference metrics which evaluate the pansharpening performance at reduced resolution according to Wald's protocol [52], we also employ the *quality with no reference* (QNR) [53] for the full-resolution assessment of the Los Angeles data set. QNR is defined as

$$\mathrm{QNR} = (1 - D_\lambda)(1 - D_S) \tag{24}$$

where D_λ and D_S are calculated to measure the spectral distortion and the spatial distortion, respectively. The smaller the D_λ and D_S, the better the quality of the pansharpened image. Thus, the optimal value of QNR is 1.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:38:43 UTC from IEEE Xplore. Restrictions apply.

TABLE I

PERFORMANCE COMPARISON OF DIFFERENT 3-D CONVOLUTION TYPES ON THE WASHINGTON DC MALL DATA SET

Type	Standard 3DConv	Separable 3DConv
CC (↑)	0.9669	**0.9718**
SAM (↓)	3.5838	**3.4106**
RMSE (↓)	0.0167	**0.0166**
ERGAS (↓)	3.6625	**3.2873**
Parameters (↓)	852639	**837365**

TABLE II

PERFORMANCE COMPARISON OF DIFFERENT FM UNIT STRUCTURES ON THE WASHINGTON DC MALL DATA SET

Structure	CC (↑)	SAM (↓)	RMSE (↓)	ERGAS (↓)
FM Unit	0.9662	3.9666	0.0184	3.6881
AFM Unit	**0.9718**	**3.4106**	**0.0166**	**3.2873**

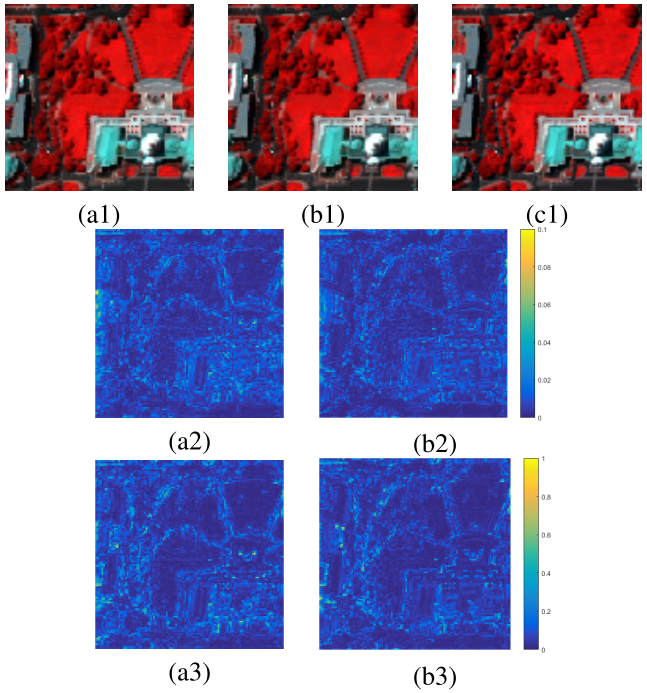


Fig. 7. Visual comparison of our method with different FM unit structures on *Washington_Patch19* (the 19th patch from the Washington DC Mall data set). First row (a1-b1): pansharpening results. Second row (a2-b2): the corresponding absolute error maps. Third row (a3-b3): the corresponding spectral angle mappings. (a1-a3) Proposed method with FM units. (b1-b3) Proposed method with AFM units. (c1) GT.

C. Model Analysis

In this section, the contributions of different components and designs in our AFM-DIN are investigated by ablation experiments, including the Sep 3DConv unit, the AFM unit, and the number of FM block.

1) Sep 3DConv Unit: To validate the efficiency of our Sep 3DConv units, we conduct a set of experiments on the Washington DC Mall data set, where the standard 3DConv layers are introduced to replace the Sep 3DConv units for comparison. Specifically, the filter sizes in the standard 3DConv layers are set to $3 \times 3 \times 3$, $5 \times 5 \times 5$, and $7 \times 7 \times 7$ for multiscale feature extraction. The quantitative performance for different 3DConv types is shown in Table I, where the better results are marked in bold. Obviously, our AFM-DIN with the Sep 3DConv units achieves superior performance in terms of all the evaluation metrics. The performance decline of using standard 3DConv layers is mainly due to the inevitable increase in model complexity, which makes the network training more difficult. On the contrary, the number of network parameters is reduced when adopting the Sep 3DConv units. The comparison results demonstrate that our Sep 3DConv units can capture multiscale HS features more efficiently.

2) AFM Unit: As illustrated in Fig. 5, the gating mechanism is deployed in each AFM unit to realize important detail information preservation and adaptive FM. To investigate the contributions of the gating mechanism in the proposed AFM unit, we construct an FM unit as the baseline structure for comparison, in which the two sigmoid layers are removed compared with the AFM unit. The quantitative evaluation results produced by our method with different structures of FM units on the Washington DC Mall data set are reported in Table II, where the better results are in bold type. It can be seen from the comparison results that the network with our AFM units outperforms that with the FM units by a larger CC value and smaller SAM, RMSE, and ERGAS values. This indicates that introducing the gating mechanism in the AFM unit not only contributes to effective incorporation of the spatial details, but also alleviates the spectral distortion. In addition, an example of visual comparison results is given to illustrate the effectiveness of the proposed AFM unit, as shown in Fig. 7. These comparisons prove that our method with the AFM units performs better in both spatial enhancement and spectral preservation.

3) Number of FM Block: The depth of our proposed AFM-DIN is mainly determined by the number of FM blocks K. To study that how many FM blocks is advantageous for pansharpening, we test the effect of K ranging from 1 to 6 on the ERGAS performance, which is a global evaluation metric. The results on the three simulated data sets are displayed in Fig. 8. We can notice that a deeper network does not always yield a better ERGAS performance. For example, the ERGAS values on the Pavia Center data set decrease at first and then increase when K changes from 2 to 4. For the Botswana data set, the values of ERGAS remain basically stable and reach the lowest when K is 4. For the Washington DC Mall data set, the performance of ERGAS varies relatively greatly and is the best when K equals 4. Based on the above observations, we set the number of FM blocks to 4 for the Botswana and Washington DC Mall data sets. For the Pavia Center data set, we adopt $K = 3$ in the following comparison experiments. For the Los Angeles data set, we empirically use the same setting of $K = 4$ as it is for the Botswana data set.

D. Comparisons With the State-of-the-Arts

To verify the effectiveness of the proposed AFM-DIN, in this section, we compare our AFM-DIN with several state-of-the-art pansharpening methods, including both classical methods such as GSA [13], PCA [14], HPF [15], and MGH [18] and emerging approaches such as DiCNN1 [23], HyperPNN2 [26], deep residual spatial attention

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:38:43 UTC from IEEE Xplore. Restrictions apply.

TABLE III
QUANTITATIVE RESULTS IN THE FORM OF MEAN±STANDARD DEVIATION OBTAINED BY DIFFERENT METHODS ON THE BOTSWANA DATA SET. BEST RESULTS ARE IN BOLD TYPE

Method	CC (↑)	SAM (↓)	RMSE (↓)	ERGAS (↓)
GSA	0.9506±0.0030	1.8123±0.2598	0.0127±0.0020	1.6476±0.1762
PCA	0.9532±0.0034	1.9356±0.3346	0.0147±0.0024	1.7632±0.2101
HPF	0.9395±0.0032	1.8707±0.2771	0.0153±0.0019	1.8278±0.1577
MGH	0.9535±0.0035	1.8417±0.2646	0.0126±0.0020	1.5738±0.1616
DiCNN1	0.9489±0.0036	1.7762±0.2057	0.0130±0.0018	1.6468±0.1428
HyperPNN2	0.9585±0.0042	1.5861±0.1253	0.0107±0.0010	1.8042±0.1097
DRSAN	0.9408±0.0052	1.7189±0.1951	0.0133±0.0012	1.8795±0.1552
DARN	0.9594±0.0051	1.5576±0.1398	0.0098±0.0010	1.5381±0.1296
DMDNet	0.9452±0.0035	1.8807±0.2425	0.0143±0.0016	1.6987±0.1655
Fusion-Net	0.9509±0.0078	1.5950±0.1582	0.0099±0.0010	1.7718±0.2379
AFM-DIN	**0.9685±0.0013**	**1.5481±0.1249**	**0.0096±0.0009**	**1.3847±0.1083**

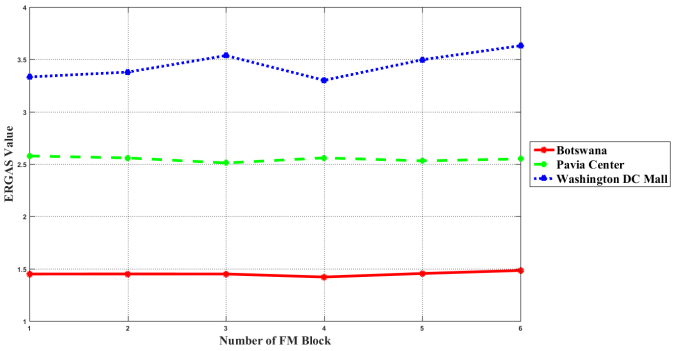


Fig. 8. ERGAS values with different numbers of FM blocks.

network (DRSAN) [28], DARN [29], DMDNet [22], and Fusion-Net [25]. Note that DiCNN1 and HyperPNN2 are adopted for comparison due to their empirically observed higher pansharpening performance than DiCNN2 and HyperPNN1. To eliminate the influence of sampling, all the methods are tested five times with randomly selected training samples. The quantitative results of five runs are averaged, and the mean and standard deviation values of all the metrics are calculated and recorded. Both the quantitative and visual evaluation results conducted on the four experimental data sets are reported as follows.

1) Results on the Botswana Data Set: Table III lists the results of quantitative evaluations obtained by different methods on the Botswana data set, where the best results are highlighted in bold type. Overall, the CNN-based methods achieve better spectral quality than the CS and MRA approaches in terms of smaller SAM values. This can be attributed to the potential of CNN in modeling the nonlinear mapping between LR and HR images. It is clear from the table that the proposed AFM-DIN outperforms the other competing methods with the highest CC score and the lowest SAM, RMSE, and ERGAS values.

Fig. 9 shows the visual results obtained by different pansharpening methods on *Botswana_Patch17*, which is the 17th patch from the Botswana data set. To observe the pansharpening results more clearly, one green-marked area is enlarged in each subimage. Through comparison between the pansharpening results and the GT image, most of the CNN-based methods exhibit less color difference than the classical methods in the enlarged regions, which indicates the better performance of CNN-based models. The visual result of the proposed AFM-DIN method shows less spatial distortion when compared with the other CNN-based methods and is the closest to the GT image.

For clarity, we display the corresponding absolute error maps of Fig. 9 in Fig. 10. As can be observed from Fig. 10, the results generated by CS and MRA approaches such as PCA, HPF and MGH suffer from serious distortions. Among the CNN-based methods, the proposed AFM-DIN achieves the least error at edges and corners of the image, which validates the excellent capacity of our AFM-DIN method in reconstructing the details.

2) Results on the Pavia Center Data Set: The quantitative results of the competing algorithms on the Pavia Center data set are shown in Table IV. Compared with the classical CS and MRA methods, the CNN-based approaches can provide higher quality of pansharpening results regarding CC, SAM, RMSE, and ERGAS performance. By observing the evaluation results of different CNN-based methods, the HyperPNN2, the DARN, and the Fusion-Net provide similar pansharpening performance on the whole. The AFM-DIN exhibits the best values on each metric, which indicates the superior properties of our proposed method in spatial details' reconstruction and spectral fidelity preservation.

To give a more intuitive comparison, we visualize the pansharpening results of *Pavia_Patch19* produced by different methods in Fig. 11, which is the 19th patch from the Pavia Center data set. Local enlargements are presented in each subfigure for easier comparison. It can be obviously seen that most CS and MRA methods suffer from spatial and color distortions in the roof regions. Compared with the GT image, the results provided by DiCNN1, HyperPNN2, DRSAN, DARN, and DMDNet have certain degrees of spatial blur, as shown in the green rectangles. To further compare the pansharpening performance visually, we also display the corresponding absolute error maps in Fig. 12. Considering the results of both Figs. 11 and 12, we can observe that our AFM-DIN method obtains the most satisfactory pansharpened image and the darkest error map. This observation is in agreement with the quantitative results reported in Table IV.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:38:43 UTC from IEEE Xplore. Restrictions apply.

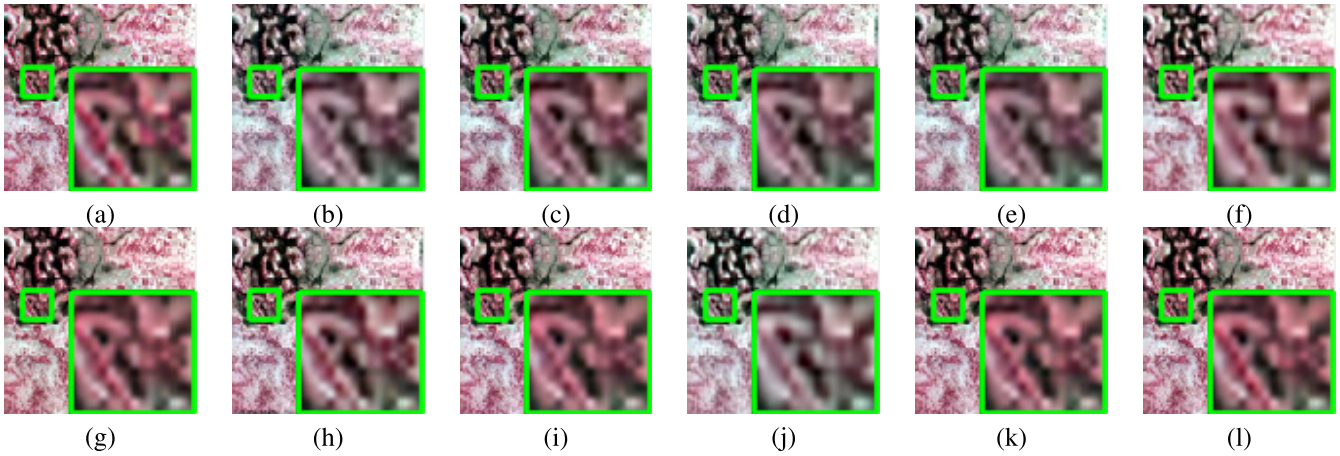


Fig. 9. Visual results obtained by different methods on *Botswana_Patch17* from the Botswana data set. (a) GT. (b) GSA. (c) PCA. (d) HPF. (e) MGH. (f) DiCNN1. (g) HyperPNN2. (h) DRSAN. (i) DARN. (j) DMDNet. (k) Fusion-Net. (l) AFM-DIN.

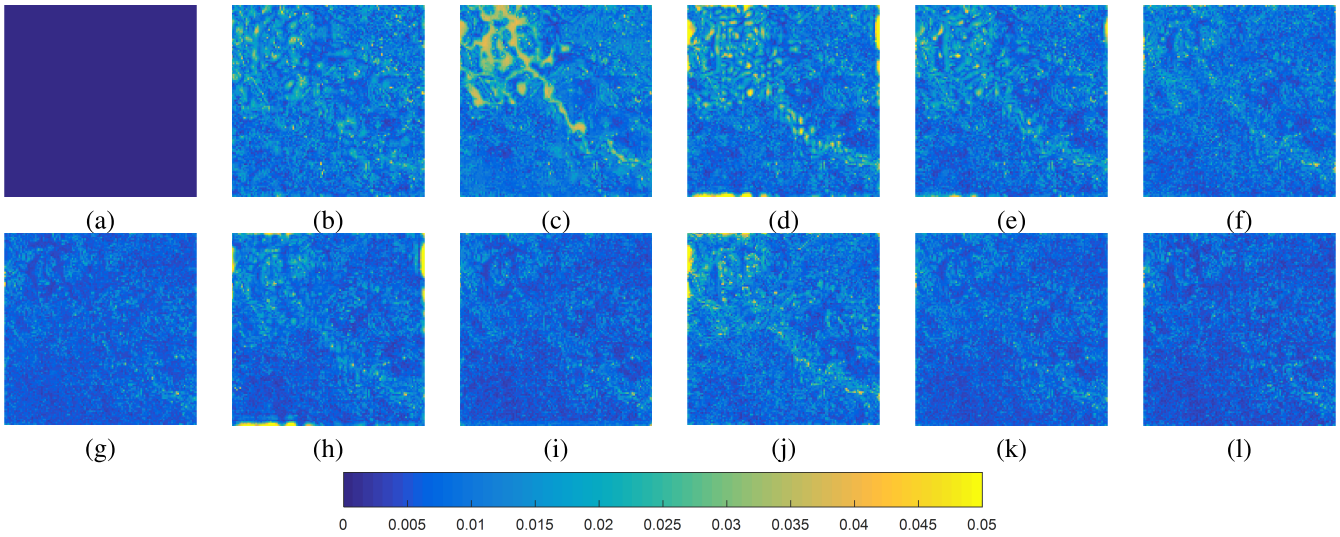


Fig. 10. Absolute error maps of Fig. 9. (a) GT. (b) GSA. (c) PCA. (d) HPF. (e) MGH. (f) DiCNN1. (g) HyperPNN2. (h) DRSAN. (i) DARN. (j) DMDNet. (k) Fusion-Net. (l) AFM-DIN.

TABLE IV

QUANTITATIVE RESULTS IN THE FORM OF MEAN±STANDARD DEVIATION OBTAINED BY DIFFERENT METHODS ON THE PAVIA CENTER DATA SET. BEST RESULTS ARE IN BOLD TYPE

Method	CC (↑)	SAM (↓)	RMSE (↓)	ERGAS (↓)
GSA	0.9498±0.0059	7.2422±0.3750	0.0294±0.0009	4.8076±0.4488
PCA	0.8956±0.0079	8.4715±0.8171	0.0432±0.0011	7.5167±0.8703
HPF	0.9425±0.0054	6.8890±0.4849	0.0314±0.0008	5.2962±0.5893
MGH	0.9499±0.0041	6.7629±0.5813	0.0307±0.0013	5.0716±0.8941
DiCNN1	0.9786±0.0022	5.0370±0.5337	0.0180±0.0007	3.0502±0.3515
HyperPNN2	0.9798±0.0015	4.8821±0.5642	0.0171±0.0008	2.9725±0.4192
DRSAN	0.9644±0.0045	6.4086±0.5168	0.0248±0.0012	4.0247±0.4588
DARN	0.9812±0.0019	4.7176±0.5872	0.0165±0.0010	2.8675±0.3777
DMDNet	0.9645±0.0037	6.0791±0.4099	0.0235±0.0007	4.0357±0.4103
Fusion-Net	0.9797±0.0018	4.7458±0.5289	0.0175±0.0008	2.9675±0.3852
AFM-DIN	**0.9815±0.0010**	**4.0584±0.2012**	**0.0157±0.0005**	**2.5035±0.2429**

To illustrate the advantage of the proposed AFM-DIN method in spectral information preservation, we plot the spectral difference curves of different methods at two randomly selected pixels on *Pavia_Patch19* in Fig. 13. The curves reflect the variation of spectral difference values between the GT image and the pansharpening results. The closer the curve is to the dotted line, the better the spectral fidelity. As seen in Fig. 13, our AFM-DIN method achieves the closest approximation to the dotted line for both pixels, especially in the last dozens of spectral

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:38:43 UTC from IEEE Xplore. Restrictions apply.

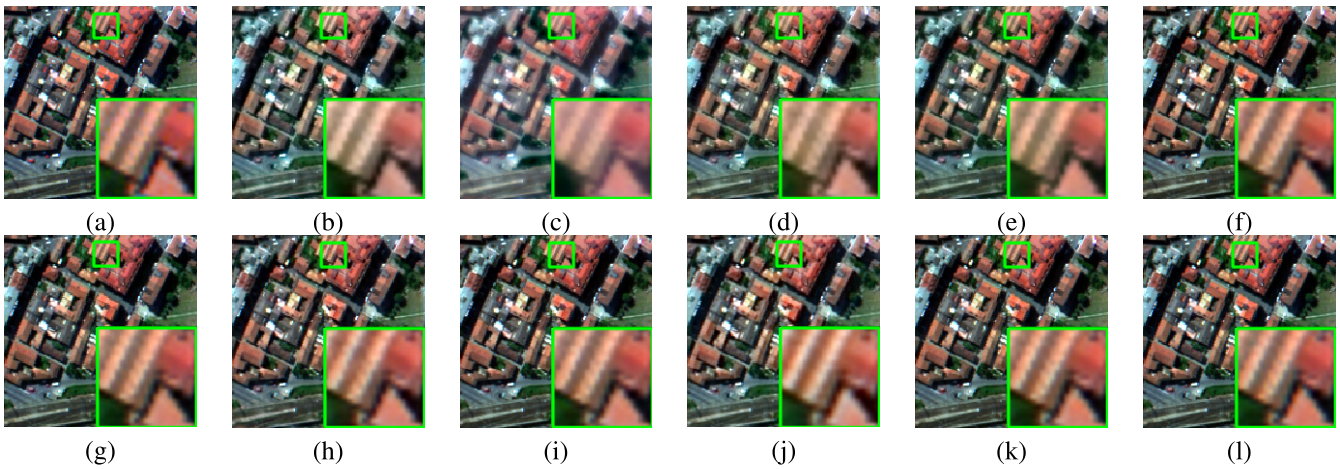


Fig. 11. Visual results obtained by different methods on *Pavia_Patch19* from the Pavia Center data set. (a) GT. (b) GSA. (c) PCA. (d) HPF. (e) MGH. (f) DiCNN1. (g) HyperPNN2. (h) DRSAN. (i) DARN. (j) DMDNet. (k) Fusion-Net. (l) AFM-DIN.

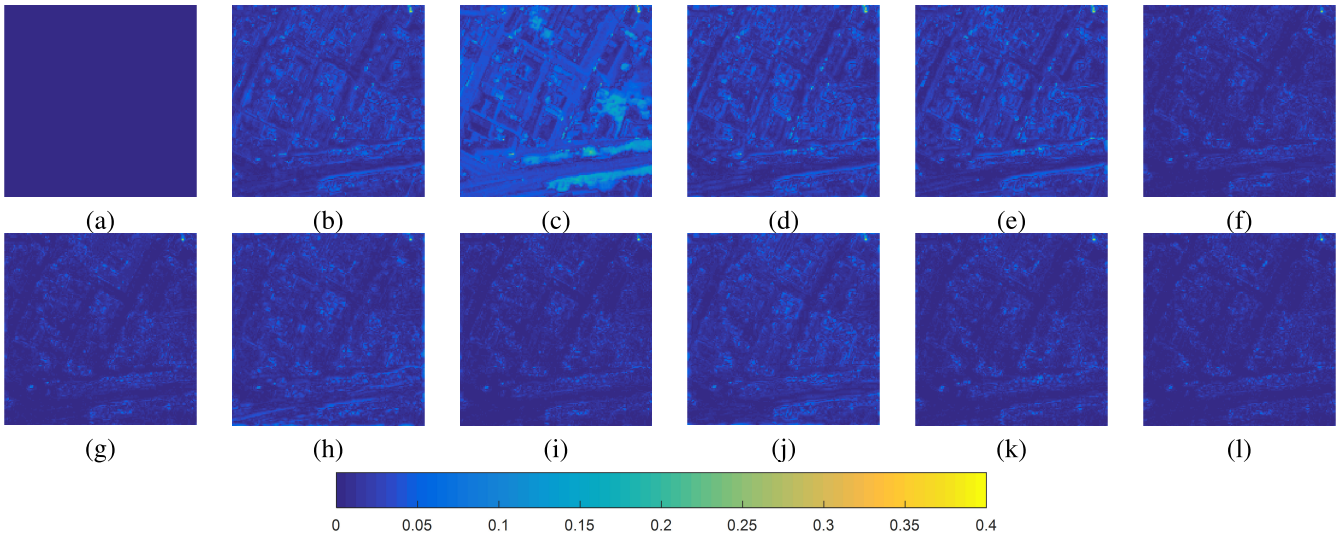


Fig. 12. Absolute error maps of Fig. 11. (a) GT. (b) GSA. (c) PCA. (d) HPF. (e) MGH. (f) DiCNN1. (g) HyperPNN2. (h) DRSAN. (i) DARN. (j) DMDNet. (k) Fusion-Net. (l) AFM-DIN.

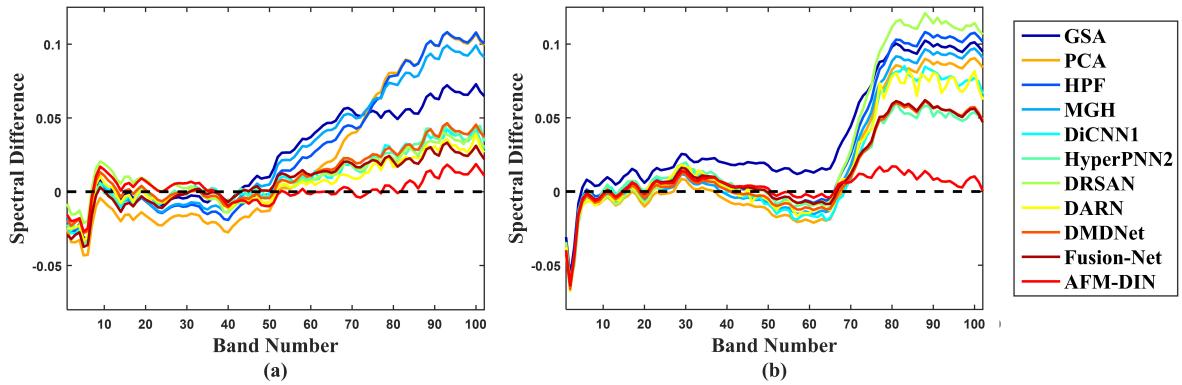


Fig. 13. Spectral difference curves of different methods at two randomly selected pixels on *Pavia_Patch19*. (a) Pixel (61, 43). (b) Pixel (131, 91).

bands. This further demonstrates the superior performance of the proposed AFM-DIN method in preserving spectral characteristics.

3) Results on the Washington DC Mall Data Set: Table V summarizes the quantitative evaluation results of pansharpening performance on the Washington DC Mall data set. By observing all the values of different metrics, we can see that our AFM-DIN approach achieves the best performance, particularly in terms of CC and ERGAS results. Although the HyperPNN2 method obtains the closest SAM and RMSE performance to ours, the AFM-DIN still offers noticeable improvements on CC and ERGAS values over HyperPNN2. This is due to the fact that our proposed AFM-DIN model enables extracting and injecting the spatial details

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:38:43 UTC from IEEE Xplore. Restrictions apply.

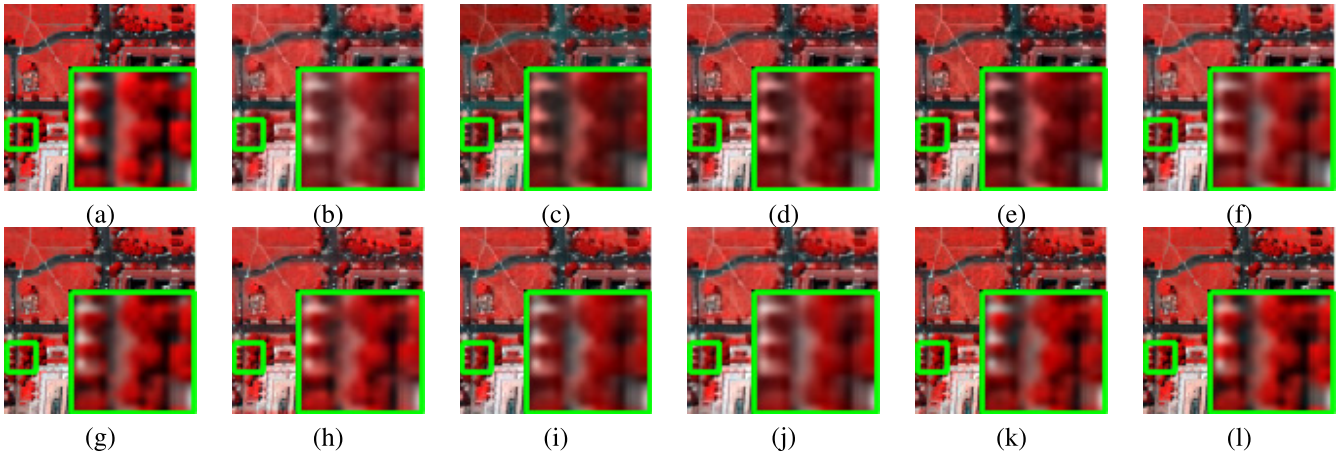


Fig. 14. Visual results obtained by different methods on *Washington_Patch10* from the Washington DC Mall data set. (a) GT. (b) GSA. (c) PCA. (d) HPF. (e) MGH. (f) DiCNN1. (g) HyperPNN2. (h) DRSAN. (i) DARN. (j) DMDNet. (k) Fusion-Net. (l) AFM-DIN.

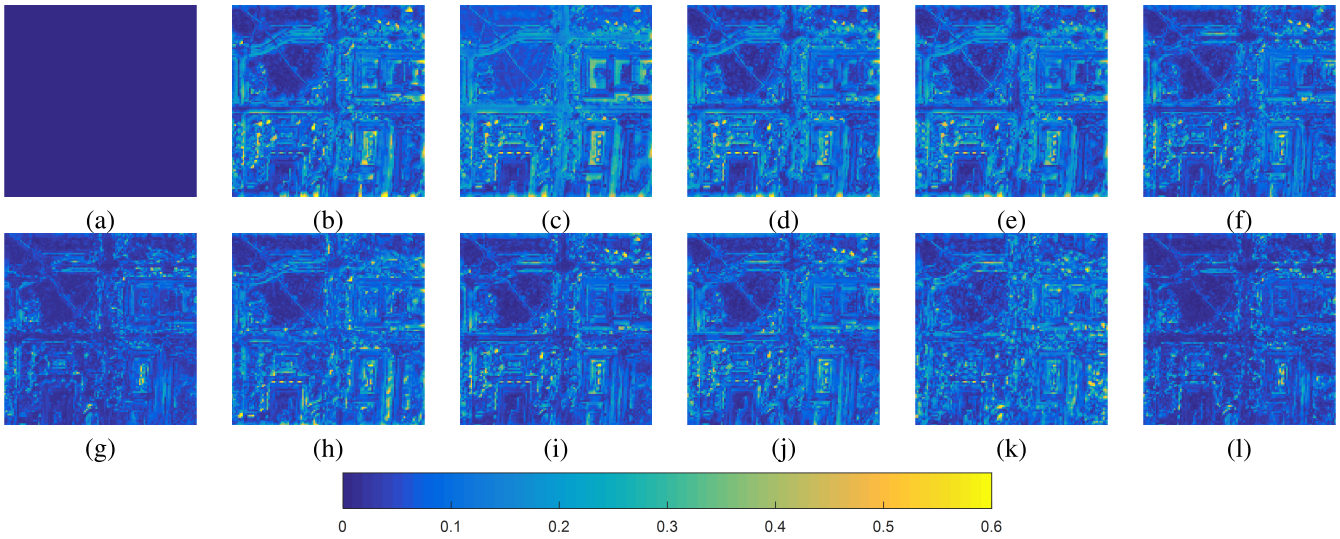


Fig. 15. Spectral angle mappings of Fig. 14. (a) GT. (b) GSA. (c) PCA. (d) HPF. (e) MGH. (f) DiCNN1. (g) HyperPNN2. (h) DRSAN. (i) DARN. (j) DMDNet. (k) Fusion-Net. (l) AFM-DIN.

TABLE V

QUANTITATIVE RESULTS IN THE FORM OF MEAN±STANDARD DEVIATION OBTAINED BY DIFFERENT METHODS ON THE WASHINGTON DC MALL DATA SET. BEST RESULTS ARE IN BOLD TYPE

Method	CC (↑)	SAM (↓)	RMSE (↓)	ERGAS (↓)
GSA	0.9361±0.0013	6.2675±0.0755	0.0278±0.0008	4.7366±0.0273
PCA	0.8811±0.0177	7.9007±0.1461	0.0390±0.0031	6.7037±0.3206
HPF	0.9300±0.0018	6.1357±0.0341	0.0291±0.0008	4.9869±0.0283
MGH	0.9416±0.0020	6.0681±0.0443	0.0277±0.0008	4.5206±0.0349
DiCNN1	0.9251±0.0110	4.9031±0.1301	0.0223±**0.0003**	6.3727±1.1762
HyperPNN2	0.9382±0.0014	3.5239±0.0831	0.0168±0.0004	6.5548±1.4304
DRSAN	0.8941±0.0023	5.5498±0.1284	0.0241±0.0004	9.1377±0.8305
DARN	0.9197±0.0031	4.9747±0.2324	0.0217±0.0007	7.2830±0.2741
DMDNet	0.9435±0.0012	5.2906±0.0388	0.0242±0.0007	4.5329±0.0452
Fusion-Net	0.9180±0.0057	4.7919±0.9706	0.0209±0.0032	8.5265±1.7692
AFM-DIN	**0.9718±0.0009**	**3.4106±0.0334**	**0.0166±0.0003**	**3.2873±0.0257**

effectively, while better preserving the spectral information simultaneously.

In Figs. 14 and 15, we display the pansharpened images and the corresponding spectral angle mappings obtained by all the competing methods on the 10th patch from the Washington DC Mall data set (called *Washington_Patch10*) for visual inspection. The selected areas are magnified for clearer observations in Fig. 14. It can be observed that there are obvious color differences and spatial distortions in the enlarged areas of the results generated by CS and MRA methods. Compared with the other CNN-based pansharpening methods, our proposed AFM-DIN presents more accurate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:38:43 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

QUANTITATIVE RESULTS IN THE FORM OF MEAN±STANDARD DEVIATION OBTAINED BY DIFFERENT METHODS ON THE LOS ANGELES DATA SET. BEST RESULTS ARE IN BOLD TYPE

Method	Reduced-resolution				Full-resolution		
	CC (↑)	SAM (↓)	RMSE (↓)	ERGAS (↓)	D_λ (↓)	D_S (↓)	QNR (↑)
GSA	0.9340±0.0020	8.7066±0.2006	0.0396±0.0006	8.0211±0.0145	0.0434±0.0005	0.1705±0.0025	0.7939±0.0012
PCA	0.9127±0.0012	8.9861±0.0650	0.0459±0.0004	9.2143±0.1057	0.0395±0.0012	0.2742±0.0027	0.6973±0.0016
HPF	0.9098±0.0031	9.3873±0.0241	0.0451±0.0003	9.1266±0.0166	0.0389±0.0050	0.0794±0.0053	0.8895±0.0071
MGH	0.9443±0.0016	8.2220±0.0209	0.0362±**0.0002**	7.3415±0.0652	0.0545±0.0015	0.1226±0.0022	0.8293±0.0012
DiCNN1	0.9752±0.0013	6.8277±0.0287	0.0233±0.0004	4.6773±0.0596	0.0384±0.0015	0.0312±0.0020	0.9291±0.0051
HyperPNN2	0.9769±0.0026	7.4108±0.0424	0.0230±0.0005	4.4314±0.1015	0.0869±0.0011	0.0607±0.0028	0.8594±0.0012
DRSAN	0.9688±0.0015	7.9731±0.0950	0.0266±**0.0002**	5.2601±0.0164	0.0460±**0.0004**	0.0319±0.0024	0.9282±0.0013
DARN	0.9764±0.0066	6.9832±0.0146	0.0225±0.0008	4.3560±0.0982	0.0403±0.0009	0.0399±0.0039	0.9218±0.0047
DMDNet	0.9708±0.0011	7.2452±0.0238	0.0247±0.0005	4.9486±0.0702	0.0486±0.0014	0.0308±0.0018	0.9272±0.0015
Fusion-Net	0.9767±0.0012	7.0598±0.0147	0.0224±0.0005	4.4416±0.1108	0.0470±0.0007	0.0348±0.0031	0.9202±0.0024
AFM-DIN	**0.9773±0.0010**	**6.7976±0.0127**	**0.0221±0.0002**	**4.3402±0.0135**	**0.0373±0.0004**	**0.0304±0.0017**	**0.9336±0.0011**

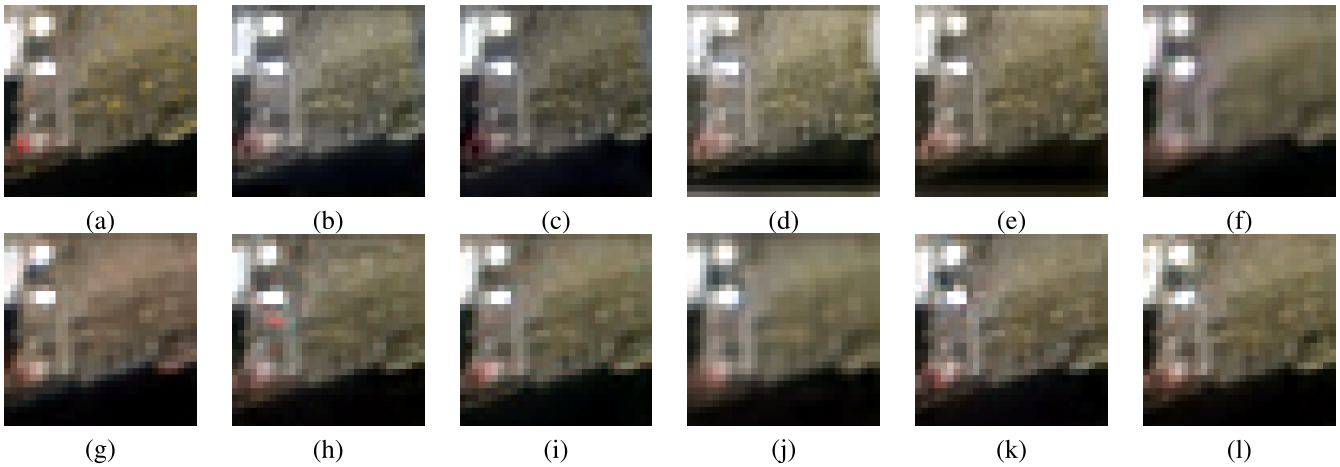


Fig. 16. Reduced-resolution pansharpening results obtained by different methods on *LosAngeles_Patch6* from the Los Angeles data set. (a) GT. (b) GSA. (c) PCA. (d) HPF. (e) MGH. (f) DiCNN1. (g) HyperPNN2. (h) DRSAN. (i) DARN. (j) DMDNet. (k) Fusion-Net. (l) AFM-DIN.

local structures and less spectral distortions with the GT, thus obtaining the very dark spectral angle mapping in Fig. 15. This observation is consistent with the objective evaluation results shown in Table V. These comparisons manifest the superiority of our proposed AFM-DIN method in both spatial quality enhancement and spectral information maintenance.

4) Results on the Los Angeles Data Set: The quantitative results of different methods at both reduced and full resolutions on the Los Angeles data set are reported in Table VI. It is clear to see that the CNN-based methods can obtain significant performance gains than traditional CS and MRA methods with respect to all the full-reference metrics. This demonstrates the powerful representation ability of CNN models in HS pansharpening. Furthermore, the proposed AFM-DIN outperforms the other six CNN-based methods for all the metrics. Under the full-resolution measures, the situation is slightly different. For example, although the results of HPF are worse than that of CNN-based approaches at reduced-resolution assessments, HPF exhibits a better QNR performance than HyperPNN2. The relatively poor spatial quality of CS and MRA methods can be seen from the higher values of D_S. Similar to the analysis at reduced resolution, the best full-resolution evaluation results are still achieved by our AFM-DIN, which testifies the robustness of the proposed AFM-DIN in pansharpening of real HS data.

Fig. 16 shows the reduced-resolution pansharpening results obtained by different methods on the sixth patch from the Los Angeles data set, here named *LosAngeles_Patch6*. It can be seen that the results of GSA and PCA methods suffer from severe spectral distortion. The pansharpened image of HPF has obvious artifacts. The MGH obtains a better visual result among the CS and MRA methods. For the CNN-based approaches, the DiCNN1 and DMDNet fail to reconstruct spatial textures. The color in the results of HyperPNN2 and DRSAN differs from that in the GT image greatly. Besides, there exists noticeable loss of spectral information in the bottom-left part of the pansharpened images produced by DARN and Fusion-Net methods. By contrast, the result of our AFM-DIN method exhibits better performance in spatial restoration and spectral fidelity, which is closest to the GT image.

Fig. 17 shows the visual results obtained through full-resolution pansharpening on *LosAngeles_Patch6*. Since there is no GT image, we present the bicubic upsampled HSI as a reference in Fig. 17(a). Overall, benefiting from the details introduced by the HR-PAN image, most pansharpening results exhibit better visual performance compared to the upsampled HSI. However, there is obvious spectral distortion in the image reconstructed by PCA. The results of HPF, MGH, DiCNN1, HyperPNN2, DRSAN, and DMDNet methods suffer

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:38:43 UTC from IEEE Xplore. Restrictions apply.

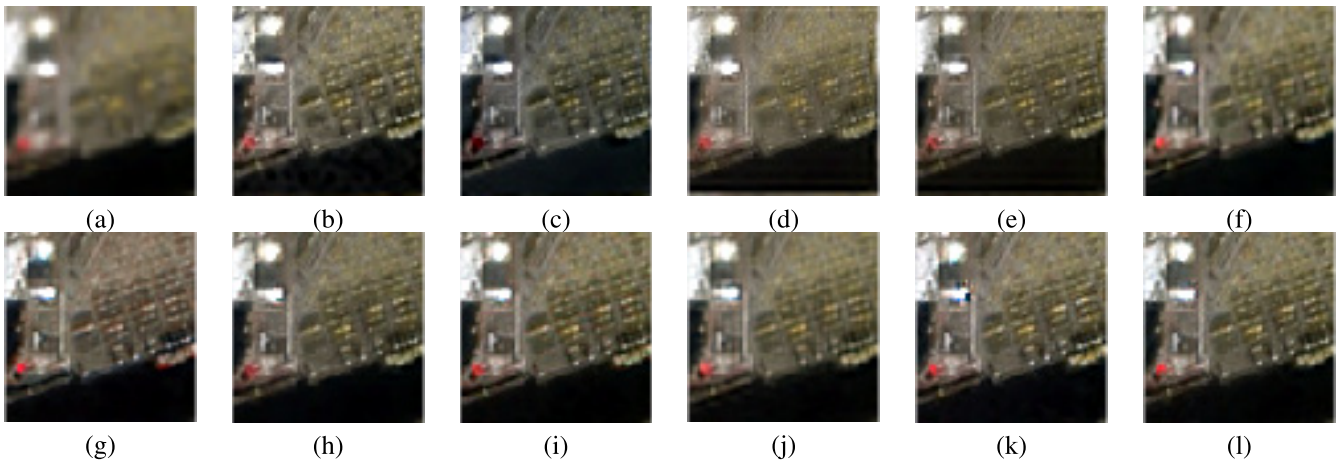


Fig. 17. Full-resolution pansharpening results obtained by different methods on *LosAngeles_Patch6* from the Los Angeles data set. (a) Upsampled HSI. (b) GSA. (c) PCA. (d) HPF. (e) MGH. (f) DiCNN1. (g) HyperPNN2. (h) DRSAN. (i) DARN. (j) DMDNet. (k) Fusion-Net. (l) AFM-DIN.

TABLE VII

PARAMETERS AND COMPUTATIONAL TIMES OF DIFFERENT CNN-BASED METHODS ON THREE SIMULATED DATASETS. BEST RESULTS ARE IN BOLD TYPE

Method	Botswana		Pavia Center		Washington DC Mall	
	Parameters (M)	Training/Testing Time (s)	Parameters (M)	Training/Testing Time (s)	Parameters (M)	Training/Testing Time (s)
DiCNN1	0.2048	**537**/0.0219	0.1552	**1393/0.0251**	0.2578	**730**/0.0281
HyperPNN2	**0.1384**	1080/0.0205	0.1329	1963/0.0288	**0.1444**	1284/0.0277
DRSAN	0.4624	738/**0.0195**	0.4128	1538/0.0273	0.5154	1012/0.0256
DARN	0.4660	817/0.0201	0.4164	1773/0.0270	0.5190	955/0.0319
DMDNet	0.2026	1815/0.0277	0.1778	4320/0.0329	0.2291	1883/0.0367
Fusion-Net	0.1574	953/0.0202	**0.1326**	2599/0.0278	0.1840	1184/**0.0255**
AFM-DIN	0.6827	4260/0.0590	0.5382	8900/0.0742	0.8374	5490/0.0830

from varying degrees of spatial blurring, which can be easily observed from the top-right corner of the reconstructed images. By comparison, with more clear spatial details and negligible color difference, the pansharpened image of the proposed AFM-DIN method appears to be more satisfactory, thus demonstrating the validity and superiority of our AFM-DIN model in solving the real-world pansharpening problem. This is because our AFM-DIN method enables both effective extraction and proper injection of details, which is extremely important for HS pansharpening.

E. Model Complexity Analysis

For CNN-based pansharpening methods, network parameters and computational time are important indicators for the analysis of model complexity. To evaluate the complexity quantitatively, we provide the number of parameters and the computational times for both training and testing of different CNN-based methods on three simulated datasets in Table VII. The experiments of each network are conducted on the same GTX 1080Ti GPU.

It can be seen from Table VII that the number of parameters of our AFM-DIN is larger than that of the other CNN-based models. This is mainly due to the utilization of 3-D convolutions and the design of deeper network in the AFM-DIN. With respect to the computational times, we can notice that the training stages of all the CNN-based methods take a relatively long time. However, once the networks are well trained, the testing times of these approaches are very small. Moreover, the testing time of our AFM-DIN is very competitive, which just slightly increases in comparison with the other CNN-based approaches. This is thanks to the strong computing power of the GPU, which effectively accelerates the processing of pansharpening and greatly reduces the inference time. Taking the pansharpening performance into consideration, the proposed AFM-DIN method can significantly boost the pansharpening accuracy in an acceptable computational complexity. Our future work will focus on optimizing the network structure to further decrease the complexity while ensuring the accuracy of HS pansharpening.

V. CONCLUSION

In this article, we present an effective HS pansharpening method called AFM-DIN, which mainly consists of four phases including high-frequency details generation, multiscale feature extraction, adaptive FM, and high-resolution image reconstruction. First, we employ a novel OctConv unit to decompose the HR-PAN image into high- and low-frequency feature maps, and then combine the high-frequency ones together to generate the comprehensive PAN detail information. Second, we creatively design the spatial and spectral Sep 3DConv units with multiple kernel sizes to capture multiscale HS features efficiently. Third, we propose a new AFM module to realize effective injection of details, where several FM blocks equipped with AFM units are cascaded. In particular, the elaborately designed AFM unit with the gating mechanism enables not only to incorporate the crucial PAN detail information effectively, but also to alleviate the spectral distortion

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:38:43 UTC from IEEE Xplore. Restrictions apply.

of the pansharpened image adaptively. Finally, the desired HR-HSI is produced by adding the details reconstructed from modulated features to the upsampled HSI. Through extensive experiments and analysis on simulated and real HS data sets, we conclude that our proposed AFM-DIN outperforms the state-of-the-art pansharpening methods in terms of both quantitative and visual evaluations.

REFERENCES

[1] J. Li, Y. Li, R. Song, S. Mei, and Q. Du, "Local spectral similarity preserving regularized robust sparse hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7756–7769, Oct. 2019.

[2] F. Luo, B. Du, L. Zhang, L. Zhang, and D. Tao, "Feature learning using spatial–spectral Hypergraph discriminant analysis for hyperspectral image," *IEEE Trans. Cybern.*, vol. 49, no. 7, pp. 2406–2419, Jul. 2019.

[3] S. Mei, X. Chen, Y. Zhang, J. Li, and A. Plaza, "Accelerating convolutional neural network-based hyperspectral image classification by step activation quantization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.

[4] B. Xi, J. Li, Y. Li, R. Song, D. Hong, and J. Chanussot, "Few-shot learning with class-covariance metric for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5079–5092, 2022.

[5] Y. Shi, J. Li, Y. Zheng, B. Xi, and Y. Li, "Hyperspectral target detection with RoI feature transformation and multiscale spectral attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5071–5084, Jun. 2021.

[6] Z. Wu, W. Zhu, J. Chanussot, Y. Xu, and S. Osher, "Hyperspectral anomaly detection via global and local joint modeling of background," *IEEE Trans. Signal Process.*, vol. 67, no. 14, pp. 3858–3869, Jul. 2019.

[7] W. Dong *et al.*, "Hyperspectral image super-resolution via non-negative structured sparse representation," *IEEE Trans. Image Process.*, vol. 25, no. 5, pp. 2337–2352, May 2016.

[8] B. Datt, T. R. McVicar, T. G. Van Niel, D. L. B. Jupp, and J. S. Pearlman, "Preprocessing EO-1 Hyperion hyperspectral data to support the application of agricultural indexes," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 6, pp. 1246–1259, Jun. 2003.

[9] G. A. Licciardi, M. M. Khan, J. Chanussot, A. Montanvert, L. Condat, and C. Jutten, "Fusion of hyperspectral and panchromatic images using multiresolution analysis and nonlinear PCA band reduction," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2011, pp. 1783–1786.

[10] L. Loncan *et al.*, "Hyperspectral pansharpening: A review," *IEEE Trans. Geosci. Remote Sens.*, vol. 3, no. 3, pp. 27–46, Sep. 2015.

[11] G. Vivone *et al.*, "A new benchmark based on recent advances in multispectral pansharpening: Revisiting pansharpening with classical and emerging pansharpening methods," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 1, pp. 53–81, Mar. 2021.

[12] T.-M. Tu, S.-C. Su, H.-C. Shyu, and P. S. Huang, "A new look at IHS-like image fusion methods," *Inf. Fusion*, vol. 2, no. 3, pp. 177–186, Sep. 2001.

[13] B. Aiazzi, S. Baronti, and M. Selva, "Improving component substitution pansharpening through multivariate regression of MS + pan data," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3230–3239, Oct. 2007.

[14] V. P. Shah, N. H. Younan, and R. L. King, "An efficient pan-sharpening method via a combined adaptive PCA approach and contourlets," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 5, pp. 1323–1335, May 2008.

[15] P. S. Chavez, S. C. Slides, and J. A. Anderson, "Comparison of three different methods to merge multiresolution and multispectral data: Landsat TM and SPOT panchromatic," *Photogramm. Eng. Remote Sens.*, vol. 57, no. 3, pp. 295–303, Mar. 1991.

[16] J. G. Liu, "Smoothing filter-based intensity modulation: A spectral preserve image fusion technique for improving spatial details," *Int. J. Remote Sens.*, vol. 21, no. 18, pp. 3461–3472, Dec. 2000.

[17] B. Aiazzi, L. Alparone, S. Baronti, A. Garzelli, and M. Selva, "MTF-tailored multiscale fusion of high-resolution MS and PAN imagery," *Photogramm. Eng. Remote Sens.*, vol. 72, no. 5, pp. 591–596, May 2006.

[18] G. Vivone, R. Restaino, M. D. Mura, G. Licciardi, and J. Chanussot, "Contrast and error-based fusion schemes for multispectral image pansharpening," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 5, pp. 930–934, May 2014.

[19] G. Licciardi, G. Vivone, M. D. Mura, R. Restaino, and J. Chanussot, "Multi-resolution analysis techniques and nonlinear PCA for hybrid pansharpening applications," *Multidimensional Syst. Signal Process.*, vol. 27, no. 4, pp. 807–830, Oct. 2016.

[20] G. Masi, D. Cozzolino, L. Verdoliva, and G. Scarpa, "Pansharpening by convolutional neural networks," *Remote Sens.*, vol. 8, no. 7, p. 594, Jul. 2016.

[21] J. Yang, X. Fu, Y. Hu, Y. Huang, X. Ding, and J. Paisley, "Pannet: A deep network architecture for pan-sharpening," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 1753–1761.

[22] X. Fu, W. Wang, Y. Huang, X. Ding, and J. Paisley, "Deep multiscale detail networks for multiband spectral image sharpening," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 5, pp. 2090–2104, May 2021.

[23] L. He *et al.*, "Pansharpening via detail injection based convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 4, pp. 1188–1204, Apr. 2019.

[24] K. Li, W. Xie, Q. Du, and Y. Li, "DDLPS: Detail-based deep Laplacian pansharpening for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 8011–8025, Oct. 2019.

[25] L.-J. Deng, G. Vivone, C. Jin, and J. Chanussot, "Detail injection-based deep convolutional neural networks for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6995–7010, Aug. 2021.

[26] L. He, J. Zhu, J. Li, A. Plaza, J. Chanussot, and B. Li, "HyperPNN: Hyperspectral pansharpening via spectrally predictive convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 8, pp. 3092–3100, Aug. 2019.

[27] L. He, J. Zhu, J. Li, A. Plaza, J. Chanussot, and Z. Yu, "CNN-based hyperspectral pansharpening with arbitrary resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–21, 2022.

[28] Y. Zheng, J. Li, Y. Li, Y. Shi, and J. Qu, "Deep residual spatial attention network for hyperspectral pansharpening," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Sep. 2020, pp. 2671–2674.

[29] Y. Zheng, J. Li, Y. Li, G. Jie, X. Wu, and J. Chanussot, "Hyperspectral pansharpening using deep prior and dual attention residual network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8059–8076, Nov. 2020.

[30] W. Zhang, J. Li, and Z. Hua, "Attention-based tri-UNet for remote sensing image pan-sharpening," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3719–3732, 2021.

[31] Q. Liu *et al.*, "Hybrid attention based residual network for pansharpening," *Remote Sens.*, vol. 13, no. 10, p. 1962, May 2021.

[32] P. Guan and E. Y. Lam, "Multistage dual-attention guided fusion network for hyperspectral pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[33] Y. Qu, R. K. Baghbaderani, H. Qi, and C. Kwan, "Unsupervised pansharpening based on self-attention mechanism," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3192–3208, Apr. 2021.

[34] S. Luo, S. Zhou, Y. Feng, and J. Xie, "Pansharpening via unsupervised convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4295–4310, 2020.

[35] S. Seo *et al.*, "UPSNet: Unsupervised pan-sharpening network with registration learning between panchromatic and multi-spectral images," *IEEE Access*, vol. 8, pp. 201199–201217, 2020.

[36] J. Nie, Q. Xu, and J. Pan, "Unsupervised hyperspectral pansharpening by ratio estimation and residual attention network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[37] Y. Yang, W. Dong, S. Xiao, T. Zhang, and J. Qu, "Multi-level spatial details cross-extraction and injection network for hyperspectral pansharpening," *Opt. Lett.*, vol. 47, no. 6, pp. 1371–1374, Mar. 2022.

[38] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2016.

[39] J. Kim, J. K. Lee, and K. M. Lee, "Accurate image super-resolution using very deep convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Oct. 2016, pp. 1646–1654.

[40] W.-S. Lai, J.-B. Huang, N. Ahuja, and M.-H. Yang, "Deep Laplacian pyramid networks for fast and accurate super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Mar. 2017, pp. 624–632.

[41] Y. Chen *et al.*, "Drop an octave: Reducing spatial redundancy in convolutional neural networks with octave convolution," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 3435–3444.

[42] S. Mei, R. Jiang, X. Li, and Q. Du, "Spatial and spectral joint super-resolution using convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4590–4603, Jul. 2020.

[43] Q. Li, Q. Wang, and X. Li, "Mixed 2D/3D convolutional network for hyperspectral image super-resolution," *Remote Sens.*, vol. 12, no. 10, p. 1660, May 2020.

[44] M. M. Khan, J. Chanussot, and L. Alparone, "Pansharpening of hyperspectral images using spatial distortion optimization," in *Proc. 16th IEEE Int. Conf. Image Process. (ICIP)*, Nov. 2009, pp. 2853–2856.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:38:43 UTC from IEEE Xplore. Restrictions apply.

[45] W. Dong, T. Zhang, J. Qu, S. Xiao, J. Liang, and Y. Li, "Laplacian pyramid dense network for hyperspectral pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.
[46] S. Mei, J. Ji, Y. Geng, Z. Zhang, X. Li, and Q. Du, "Unsupervised spatial–spectral feature learning by 3D convolutional autoencoder for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6808–6820, Sep. 2019.
[47] Q. Li, Q. Wang, and X. Li, "Exploring the relationship between 2D/3D convolution for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8693–8703, Oct. 2021.
[48] S. Mei, X. Li, X. Liu, H. Cai, and Q. Du, "Hyperspectral image classification using attention-based bidirectional long short-term memory network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.
[49] S. Xie, C. Sun, J. Huang, Z. Tu, and K. Murphy, "Rethinking spatiotemporal feature learning: Speed-accuracy trade-offs in video classification," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 305–321.
[50] Y. Gao, Y. Zhang, X. Li, L. Li, A. Gao, and P. Yu, "A review of guided filter based pansharpening methods," in *Proc. 4th Int. Conf. Inf. Syst. Comput. Technol.*, Jan. 2016, pp. 44–49.
[51] E. M. Middleton *et al.*, "The earth observing one (EO-1) satellite mission: Over a decade in space," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 243–256, Apr. 2013.
[52] L. Wald, T. Ranchin, and M. Mangolini, "Fusion of satellite images of different spatial resolutions: Assessing the quality of resulting images," *Photogramm. Eng. Remote Sens.*, vol. 63, no. 6, pp. 691–699, 1997.
[53] L. Alparone, B. Aiazzi, S. Baronti, A. Garzelli, F. Nencini, and M. Selva, "Multispectral and panchromatic data fusion assessment without reference," *Photogramm. Eng. Remote Sens.*, vol. 74, no. 2, pp. 193–200, Feb. 2008.



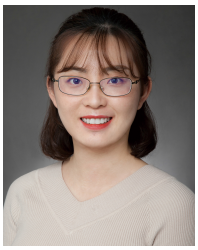
Yunsong Li (Member, IEEE) received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 1999 and 2002, respectively.

In 1999, he joined the School of Telecommunications Engineering, Xidian University, where he is currently a Professor. He is also the Director with the State Key Laboratory of Integrated Services Networks, Image Coding and Processing Center, Xidian University. His research interests include image and video processing, hyperspectral image processing, and high-performance computing.



Yuxuan Zheng (Member, IEEE) received the B.E. degree in space information and digital technology and the Ph.D. degree in information and communications engineering from the School of Telecommunications Engineering, Xidian University, Xi'an, China, in 2017 and 2022, respectively.

From 2021 to 2022, he was a Visiting Ph.D. Student with the Inria, Grenoble INP, Université Grenoble Alpes, Grenoble, France, supervised by Prof. Jocelyn Chanussot, and supported by the China Scholarship Council. He is currently a Lecturer with the State Key Laboratory of Integrated Services Networks, Xidian University. His research interests include hyperspectral remote sensing, data fusion, and deep learning.



Jiaojiao Li (Member, IEEE) received the B.E. degree in computer science and technology, the M.S. degree in software engineering, and the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 2009, 2012, and 2016, respectively.

She was an exchange Ph.D. Student with the Mississippi State University, Starkville, MS, USA, supervised by Dr. Qian Du. She is currently an Associate Professor and a Doctoral Supervisor with the State Key Laboratory of Integrated Services Networks, School of Telecommunications Engineering, Xidian University. Her research interests include hyperspectral remote sensing image analysis and processing, pattern recognition, and data compression.



Rui Song (Member, IEEE) received the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 2009.

He is currently a Professor with the State Key Laboratory of Integrated Services Networks, School of Telecommunications Engineering, Xidian University. His research interests include image and video coding algorithms, very large-scale integration (VLSI) architecture design, intelligent image processing, and understanding and reconstruction of 3-D scenes.



Jocelyn Chanussot (Fellow, IEEE) received the M.Sc. degree in electrical engineering from the Grenoble Institute of Technology (Grenoble INP), Grenoble, France, in 1995, and the Ph.D. degree from the Université de Savoie, Annecy, France, in 1998.

Since 1999, he has been with Grenoble INP, where he is currently a Professor of signal and image processing. He was a Visiting Scholar with Stanford University, Stanford, CA, USA; the KTH Royal Institute of Technology, Stockholm, Sweden; and the National University of Singapore (NUS), Singapore. Since 2013, he has been an Adjunct Professor with the University of Iceland, Reykjavik, Iceland. From 2015 to 2017, he was a Visiting Professor with the University of California at Los Angeles (UCLA), Los Angeles, CA, USA. He holds the AXA Chair in remote sensing and is an Adjunct Professor with the Chinese Academy of Sciences, Aerospace Information Research Institute, Beijing, China. His research interests include image analysis, hyperspectral remote sensing, data fusion, machine learning, and artificial intelligence.

Dr. Chanussot was a member of the Machine Learning for Signal Processing Technical Committee of the IEEE Signal Processing Society from 2006 to 2008, and the Institut Universitaire de France from 2012 to 2017. He was the Founding President of the IEEE Geoscience and Remote Sensing Society (GRSS) French Chapter from 2007 to 2010, which received the 2010 IEEE GRSS Chapter Excellence Award. He was the Vice-President of the IEEE GRSS, and the In Charge of meetings and symposia from 2017 to 2019. He was the General Chair of the first IEEE GRSS Workshop on Hyperspectral Image and Signal Processing, Evolution in Remote Sensing (WHISPERS). He was the Co-Chair and the Chair of the GRSS Data Fusion Technical Committee from 2005 to 2008 and from 2009 to 2011, respectively. He was the Program Chair of the IEEE International Workshop on Machine Learning for Signal Processing in 2009. He was the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING from 2011 to 2015. He is an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON IMAGE PROCESSING, and the PROCEEDINGS OF THE IEEE. He has been a Highly Cited Researcher (Clarivate Analytics/Thomson Reuters) since 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:38:43 UTC from IEEE Xplore. Restrictions apply.